



Leiyang Hexi Animal Manure Composting Project



Document Prepared by CTI Certification Co., Ltd.

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Summary:

CTI Certification Co., Ltd. (hereafter referred to as "CTI") has been commissioned by Climate Bridge (Shanghai) Ltd. to perform the validation and verification of project activity "Leiyang Hexi Animal Manure Composting Project" (hereafter referred to as "the project activity") and reported in the joint Validation & Verification Report.

The project activity, located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, of which the main purpose is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City.

During the 3*7 renewable crediting period, the project activity avoids methane emissions through aerobic composting of solids in animal manure. The project activity could separate approximately 52,500 tonnes of solid from animal manure and produce 30,000 tonnes of organic solid fertilizers annually. It's estimated that the project activity could achieve average annual GHG emission reductions of 57,307 tCO₂e and total GHG emission reductions of 401,149 tCO₂e during the first 7-year crediting period.

A description of the validation and verification

The approved CDM methodology AMS-III.Y. "Methane avoidance through separation of solids from wastewater or manure treatment systems" (Version 4.0) is applied to quantify the GHG removals achieved in this project. The calculation of the GHG emission reductions is carried out in a transparent and conservative manner. This project is being developed in conjunction with the validation and 1st periodic of verification.

The purpose and scope of validation and verification

The validation objective is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods a procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to VCS registry.

The objective of the verification is to have an independent review ex post determination by a VVB (Validation

and Verification Body) of the monitored GHG emission reductions that have occurred as a result of the implementation of the project activity during a defined monitoring period.

The method and criteria used for validation and verification

Validation and Verification is conducted using CTI Certification Co., Ltd (CTI) procedures in line with the requirements specified in the latest version of the VCS Validation and Verification Manual and applying auditing techniques. The validation/verification team assessed the project activity's compliance against the VCS Standard Version 4.5, the selected CDM methodologies and the joint project description and monitoring report. The project is eligible under Project Scopes 13. The validation/verification criteria followed the guidance documents provided by VCS included the following:

VCS Standard Version 4.5, VCS Program Guide Version 4.4 and the applied CDM methodologies AMS-III.Y "Methane avoidance through separation of solids from wastewater or manure treatment systems" (Version 4.0) .

The number of findings raised during validation and verification

In the course of the validation and verification, 1 Clarification Requests (CLs) and 2 Corrective Action Requests (CARs) was raised and successfully closed. No Forwarded Action Requests (FARs) was raised.

Any uncertainties associated with the validation and verification

There are no restrictions of uncertainty for both validation and verification.

Summary of the validation and verification conclusions

CTI concludes that the project activity "Leiyang Hexi Animal Manure Composting Project" in China, as described in the final version of Joint-Project-Description-Monitoring-Report, meets all relevant requirements for VCS validation and verification activity and correctly applied the methodology AMS-III.Y Version 4.0. Hence CTI is able to provide positive validation opinion as per the requirement of VCS and further certify that the GHG emission reduction from the project during the monitoring period from 28/03/2022 to 30/09/2023 amount to 88,210 tCO₂e VCUs.

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1 INTRODUCTION

1.1 Objective

Climate Bridge (Shanghai) Ltd. has commissioned the CTI to carry out the Verified Carbon Standard (VCS) validation joint with 1st periodical verification of the project "Leiyang Hexi Animal Manure Composting Project" (VCS ID 4794) for the for the monitoring period from 28/03/2022 to 30/09/2023 (both days included).

The objective of validation is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to Verra registry.

The objective of verification is to have an independent review ex-post determination by a VVB of the monitored GHG emission reductions that have occurred as a result of the project activity implemented during a defined monitoring period. The evaluation is done against the requirements of the VCS Standard Version 4.5 based on the monitoring report. In order to confirm that the GHG emission reductions sound reasonable and meet the identified criteria, the verification involves the assessment of monitoring plan conformance to VCS rules and applied methodology. Verification is a requirement and is seen as necessary to provide assurance on the generation of VCUs.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the joint Project Design and Monitoring Report to validate that (a) the project design is actual, (b) the baseline scenario is correctly defined as per the applied methodology and relate tools, (c) the project is additional, (d) the monitoring plan can be implemented and is transparent and adequate and (e) all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature. The joint PD & MR is reviewed against the criteria stated in VCS standard Version 4.5 /3/ and the approved baseline and monitoring methodology AMS-III.Y., Version 4.0. The validation was based on the requirements VCS Validation and Verification Manual and VCS Program Guide version 4.4 /4/ and VCS standards Version 4.5 /3/ applying auditing techniques. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The scope of the verification is to verify that: (a) the actual monitoring system and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan; (b) the GHG emission reductions data and express a conclusion with a reasonable level of assurance about whether the reported GHG emission reductions data are free from material misstatement; (c) the reported GHG emissions data is sufficiently supported by evidence. Verification shall ensure that the reported GHG emission reductions are complete and accurate in accordance with the applicable VCS criteria in order to be certified. Verification is conducted using CTI procedures in line with the requirements specified in the VCS Program and ISO14064-3 requirements and applying auditing techniques. The verification team assessed and determined that the implementation and operation of the project activity, and steps to report GHG emission reductions comply with the VCS rules. The verification involved a document review of relevant documentation and interview with the project participants. Verification is not meant to provide any consultancy towards the project participants; however, stated request for clarifications and/or corrective actions may have provided input for improvement of the monitoring.

1.3 Level of Assurance

CTI has assessed the claims and assumptions made in the joint Project Description&Monitoring Report, i.e., assumptions regarding the determination of baseline scenario, and additionality assessment, as well as validated the assumptions, limitations, and methods used for determining parameters and calculation of baseline emission, project emission and leakage, against the VCS Standard Version 4.5, applied methodology. It is confirmed that the assumptions made in the joint Project Description&Monitoring Report are reasonable which are in line with the applied methodology and tools as well as the requirement of VCS. Based on the validated assumptions, limitations, and methods used during the validation (see section 3.4.6 for ex ante ER calculation result), the project is expected to generate average GHG emission removal of 57,307 tCO₂e per year, and it is confirmed that outcome of future activities will be achieved as the assumptions and limitations are reasonable.

The joint validation and verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per para 3.10.1 and 4.1.10 (4) of VCS Standard Version 4.5.

The assessment was conducted to provide a reasonable level of assurance of compliance against the defined audit criteria and materiality threshold within the audit scope. Based on the audit findings the validation and verification statement reasonably assure that the GHG assertion is materially correct and is a fair representation of the GHG data and information.

1.4 Summary Description of the Project

Project title	Leiyang Hexi Animal Manure Composting Project
Project Participants	Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. (Project Owner, host country, P. R. China) Climate Bridge (Shanghai) Ltd. (China)
Location of the project	Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China. Geographic coordinates: longitude of 112°48'35.8"E, 26°20'50.0"N
Project start date	28/03/2022, the operation start date
Applied Methodology/Version	AMS-III.Y., Version 4.0
Scope/Technical Area	Sectoral scope 13: Waste handling and disposal
VCS crediting period	28/03/2022 to 27/03/2029

The project activity, located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, of which the main purpose is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from existing swine farms in Leiyang City. During the 3*7 renewable crediting period, the project activity avoids methane emissions through aerobic composting of solids in animal manure. The project activity could separate approximately 52,500 tonnes of solid from animal manure and produce 30,000 tonnes of organic solid fertilizers annually. It's estimated that the project activity could achieve average annual GHG emission reductions of 57,307 tCO₂e and total GHG emission reductions of 401,149 tCO₂e during the first 7-year crediting period.

By checking Livestock Manure Treatment Agreement /69/, List of livestock farm /67/, interviewing PP during site visit, as well as onsite investigation, it was verified the swine farms involved in the project and their identification were listed as follows:

Location	Name	Year of Establishment	Business Scope
Sandu Town	Sandu Town Zhang Hong'an Pig Farm	2018	Swine farming
Donghu Town	Donghu Wei Town Dong Sanjun Pig Farm	2018	Swine farming
Donghu Town	Donghu Beilong Heping Pig Farm	2018	Swine farming
Gongping Town	Rixing Pig Farm	2017	Swine farming
Dayi Town	Hongsheng Pig Farm	2016	Swine farming
Yuqing Township	Yuqing Shixinyuan Pig Farm	2018	Swine farming
Donghu Town	Donghu Town Xinyuan Pig Farm	2016	Swine farming
Sandu Town	Sandu Town Yang Shaowu Pig Farm	2016	Swine farming
Xiaoshui Town	Xiaoshui Enrui Family Farm	2016	Swine farming
Gongping Town	Shunhe Family Farm	2018	Swine farming
Dayi Town	Dayi Town Hongli Ecological Pig Farm	2016	Swine farming
Dayi Town	Dayi Huang Xiaoying Pig Farm	2016	Swine farming
Dayi Town	Fengping Family Farm	2017	Swine farming
Dayi Town	Dayi Town Yushi Xingwang Livestock Farm	2016	Swine farming
Donghu Town	Donghu Wei Town Hu Xiaozhan Pig Farm	2018	Swine farming

Yuqing Township	Xu Xiaohong Pig Farm	2019	Swine farming
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2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

A project specific validation and verification plan was developed to guide the validation and verification auditing process to ensure efficiency and effectiveness.

The purpose of the validation and verification is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk of auditing error is reduced to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. The validation and verification were performed through means of the following three phases in accordance with the requirement of the applied methodology, and the VCS Standard (Version 4.5) and other relevant VCS requirements.

- A desk review of the joint PD & MR and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the joint validation and verification report and statement.

CTI completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable) project details; application of methodology; estimated GHG emission reduction and removals; monitoring; safeguards; SDG contributions etc.

CTI also verified the emission reductions through assessing all factors and issues that constitute the basis for emission reductions from the project, including:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

CTI assessed that the reported information in the joint PD & MR are complete and accurate in question. This joint validation and verification report describes the findings of this assessment. The information of the assessment team has been included in below of this report.

2.2 Document Review

The joint PD & MR /1/ and supporting background documents related to the project implementation were reviewed. Documents review was conducted to ensure consistency with and identify any deviation from VCS program requirements. Desk review included an examination of the project design details,

baseline scenario, additionality, ex ante and monitoring data and parameters, and ex ante quantification of GHG emission reductions. The assessment team also used additional documentation by third parties like host party legislation, technical reports referring to the project design details, baseline scenario, additionality, monitoring or to the basic conditions and technical data.

Furthermore, the joint PD & MR /1/ and emission reduction calculation spreadsheet of monitoring period from 28/03/2022 to 30/09/2023, relevant decisions, clarifications and guidance from the Verra, other information and references relevant to the project activity have also been assessed. During the desk review,

CTI has applied standard auditing techniques to assess the quality of information provided, including a review of the data and information presented to verify their completeness; a review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and an evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

The references used in the course of this validation and verification are summarized in Appendix 3. The validation and verification were performed basing on the documents check and site inspection/measurements, refer to the section 3 of this report for the validation process detail, section 3 of this report for the verification process detail, and Appendix 3 for corresponding documents review.

According to the sectoral scopes / technical area and experiences in the sectoral or national business environment, CTI has composed a project assessment team in accordance with the appointment rules in CTI. The composition of CTI assessment team has to be approved by the CTI ensuring that the required skills are covered by the team. The four qualification levels for team members that are assigned by formal appointment rules as below:

Function	Name	Technical competence	Task Performance*
Team Leader	Feng Tian	1, 13	☒ DR ☒ SV ☒ RP ☐ TR
Technical Expert	Dai Qinghua	13, 15	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1, 3, 13, 14, 15	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

2.3 Interviews

The objective of the interview process was to solicit important information from personnel related to project and relevant to the validation and verification process. Onsite interviews and information discussions were conducted with PP. The key personnel interviewed are summarized in the table below:

Interviewed personnel	Role	Date	Organization	Subject
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Ms. Liang Xiaojiao	Office Director	21/12/2023	Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd.	Project design and operation of the project activity; Status of the project (including PPs); Project proponent and ownership; Applicability of selected methodology; Applicable laws, statutes and other regulatory frameworks and any change; Baseline of the project; Emission factors of the project; Monitoring plan; Stakeholder consultation process and its outcomes; Sustainable development contributions.
Mr. Li Jun	Project Manager	21/12/2023		
Mr. Tang Feiqing	Project Administrator	21/12/2023		
Mr. Li Shaosheng	Staff	22/12/2023	Animal Husbandry and Veterinary Management Department of Agriculture and Rural Affairs Bureau	The process and participation of the stakeholder consultation; The impact of the project activity; The complaint by local stakeholders and the implementation of the mitigation measures.
Mr. Gu Yong	Director	22/12/2023	Hongsheng Livestock Farm	
Mr. Liu Zhiyong	Director	22/12/2023	Sirui Livestock Farm	
Mr. Yan Hao	Villager	22/12/2023	Local Village	
Ms. Wang Xiaomei	Villager	22/12/2023		
Mr. Liu Dachuan	Villager	22/12/2023		
Ms. Lan Donghua	Villager	22/12/2023		
Ms. Chen Yujie	Project Manager	21/12/2023 -22/12/2023	Climate Bridge (Shanghai) Ltd.	Data collection and ER calculation.

2.4 Site Visits

The assessment team performed the on-site validation and verification (Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China) on 21/12/2023 - 22/12/2023. The interviewed personnel and objective are listed in above table.

Duration of on-site inspection: 21/12/2023 - 22/12/2023

No.	Activity performed on-site	Site location	Date	Team member
1.	Opening meeting Interview with PP Representative, local officers, local residents and Operation Staff	Project Owner's office in the project site	21/12/2023	Feng Tian Dai Qinghua
2.	On-site inspection Project site, farms, composting workshops, equipments etc., interviewing stakeholders	project site	21/12/2023	Feng Tian Dai Qinghua
3.	Documents check	Project Owner's office in the project site	21/12/2023	Feng Tian Dai Qinghua
4.	Finding Summary	Project Owner's office in the project site	22/12/2023	Feng Tian Dai Qinghua
5.	Close Meeting	Project Owner's office in the project site	22/12/2023	Feng Tian Dai Qinghua

The opening meeting of the joint validation and verification process started on 21/12/2023 at Project Owner's office in the project site (Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China), as confirmed by the assessment team the project is listed on the project pipeline with a status of under validation before the opening meeting representing the beginning of the validation process.

Furthermore, the validation and verification took place after the 30-day public comment period which begun on 22/11/2023, and CTI completed the validation and verification until after the 30-day public comment period has ended.

During the on-site visit, CTI has applied standard auditing techniques to assess the quality of information provided, the implementation and current situation of the project activity, evaluate data management, QA/QC system, project technology and equipment, training provided, and monitoring etc.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for issuance.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for CTI's positive conclusion on the joint PD & MR /1/. 1 Clarification Requests (CLs), 2 Corrective Action requests (CARs) and no Forward action request (FAR) raised by CTI. And communications were made between the Client and CTI to guarantee the transparency of the validation and verification process, the concerns raised, and responses given are summarized below in the appendix 4.

The final joint PD & MR (version 3.0 dated 06/09/2024) serves as the basis for the final assessment presented. Additional changes to the project during the validation and verification process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation and verification findings	No. of CL	No. of CAR	No. of FAR
Project Description and monitoring report	00	00	00
Description of project activity	00	00	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Clarification on applicability of methodology, tool and/or standardized baseline	00	00	00
- Demonstration of additionality	00	00	00
- Project boundary	1	00	00
- Emission reductions	00	00	00
- Monitoring plan	00	00	00

- Stakeholders consultation process	00	00	00
- Public comments	00	00	00
Others (please specify) - Matter related to double counting - for validation	00	00	00
Others (please specify) - Environmental Impacts - for validation	00	1	00
Others (please specify) - Matter related to ex post monitoring - for verification	00	1	00
ER achieved, Matter related to feeder details, breakdown and Calibration - for verification	00	00	00
Total	1	2	00

2.5.1 Forward Action Requests

None FAR was raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

The project activity, located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, of which the main purpose is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City. During the 3*7 renewable crediting period, the project activity avoids methane emissions through aerobic composting of solids in animal manure. The project activity could separate approximately 52,500 tonnes of solid from animal manure and produce 30,000 tonnes of organic solid fertilizers annually. It's estimated that the project activity could achieve average annual GHG emission reductions of 57,307 tCO₂e and total GHG emission reductions of 401,149 tCO₂e during the first 7-year crediting period.

The project activity has been developed by Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. The geographic coordinates of the project activity are 112°48'35.8"E, 26°20'50.0"N, which is confirmed by site visit. By checking Project Approval /15/, the assessment team confirmed that the project activity has been approved initially on 16/07/2019 and revised on 22/04/2020 due to productivity changes of organic fertilizers. By checking Environmental Impact Assessment (EIA) Approval /17/, the assessment team confirmed that the project activity has been approved by Leiyang City Ecological Environmental Bureau on 15/01/2021. By checking Construction Contract /23/, the assessment team confirmed that the project started construction on

28/08/2019. By checking Operation Log /34/ and site interview, the assessment team confirmed that the project activity started operation on 28/03/2022.

For the project activity, the applicability of VCS program as required in para 2.1.1 of VCS Standard (Version 4.5) is justified as below:

Criteria	Justification
The scope of the VCS Program includes: 1) The seven Kyoto Protocol greenhouse gases.	During the 3*7 renewable crediting period, the Project, via treatment of manure by composting, can reduce CH ₄ emissions by avoiding methane emissions that would be emitted to the atmosphere directly without any methane recovery and destruction facility. Therefore, this criteria is applicable.
2) Ozone-depleting substances.	This criteria is not applicable for the project activity.
3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	This criteria is not applicable for the project activity.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	The project activity applies methodology AMS-III.Y. (Version 4.0), the methodologies approved under CDM Program, which is a VCS approved GHG program.
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	This criteria is not applicable for the project activity.
The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. The VCS Program also excludes the following project activities under the circumstances indicated in Table 1 of VCS Standard (Version 4.5).	The project activity, located in Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China, of which the main purpose is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City. The project activity is not excluded under the circumstances indicated in Table 1 of VCS Standard (Version 4.5).

Project design, including eligibility criteria for grouped projects

Through document review, i.e. FSR /14/, and on site inspection, the assessment team confirmed that treatment process that applied by the project activity includes manure collection, solid separation system, and composting system. Based on the FSR /14/, onsite investigation and interviews with project owners and workers, it was verified the separated liquids after solid-liquid separation undergo deammonification, denitrification, and concentration to produce high-efficiency liquid fertilizer. Methane generated during this process is not being recovered for utilization as the project system is not

equipped with methane recovery system. By checking Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and via on site visit, the technical parameters of main equipment was verified and listed in table below.

Equipment	Type	Number	Capacity	Technical Lifetime
Solid-liquid Separation Equipment	LX-950K-3Q-20hDAF-20	1	Screening: 0.75kw Screw-pressing: 3kw	15 yrs
Compost turner	RY-ZFP40B	1	50.7kw	15 yrs
Baler	RY-B50F	1	2.2kw	15 yrs

The project activity is not a grouped project activity. Therefore, this section is not applicable.

Project proponent and other entities involved in the project

Through document review and site interview, the assessment team confirmed the details of the project proponent as:

Organization name	Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd.
Contact person	Jun Li
Title	Project Manager
Address	Group 4, Dajin Village, Caizichi Street, Leiyang City, Hengyang City, Hunan Province, People's Republic of China
Telephone	+86 2123019950
Email	3542346576@qq.com

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultant
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120
Telephone	+86-21 62462036
Email	gao.zhiwen@climatebridge.com; projects@climatebridge.com

Project Ownership

By checking Business License /28/, Project Approval /15/ and EIA Approval /17/, the assessment team confirmed that Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. is the project owner and also project proponent (PP) of project activity. Therefore,

Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. has the legal right to control and operate the project activities.

Project Start Date

By checking Operation Log /34/ of the project activity and site interview, the assessment team confirmed that the project activity started operation on 28/03/2022, which is the start date of the project activity.

Project crediting period Date

The project activity adopts 3*7 renewable crediting period. the assessment team confirms that the starting and ending dates of the first crediting period is as below:

Crediting Period Start date: 28/03/2022

Crediting Period End date: 27/03/2029

Project Scale and Estimated GHG Emission Reductions or Removals

As the estimated annual average GHG emission reductions or removal per year is 57,307 tCO₂e which is less than 300,000 tonnes of CO₂e per year, thus the project falls in the category of Project. Therefore, the assessment team confirms that the project scale falls under "Project".

Project Scale	
Project	✓
Large project	

Through checking emission reductions calculation spreadsheet provided by PP, the assessment team was able to confirm that the estimated GHG Emission Reductions or Removals of the project activity is as follows:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
28/03/2022 to 31/12/2022	43,805
2023	57,307
2024	57,307
2025	57,307
2026	57,307
2027	57,307
2028	57,307
01/01/2029 to 27/03/2029	13,502
Total estimated ERs	401,149
Total number of crediting	7

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
years	
Average annual ERs	57,307

The above estimated emission reduction is confirmed by the assessment team via emission reduction calculation spreadsheet. the assessment team confirmed that the calculation is correct and conservative.

Project location

The project activity is located at Group 1, Wufeng Village, Xiaoshui Town, Leiyang City, Hengyang City, Hunan Province, People's Republic of China. The project activity has been developed by Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd. The project is located at 112°48'35.8"E, 26°20'50.0"N, which is confirmed by site visit.

Conditions prior to project initiation

Before the implementation of the project activity, animal manure was left to decay anaerobically in uncovered anaerobic lagoons within the project boundary and was used for land application in nearby farms, the generated methane is emitted to the atmosphere.

Project compliance with applicable laws, statutes and other regulatory frameworks

The assessment team confirms that the Project has been approved by Chinese government by checking Project approval /15/ and Environmental Impact Assessment (EIA) Approval /17/.

By checking laws and regulations, i.e. Environmental Protection Law of the People's Republic of China, Environmental Impact Assessment Law of the People's Republic of China, Animal Husbandry Law of the People's Republic of China, Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution, Law of the People's Republic of China on the Prevention and Control of Air Pollution, Law of the People's Republic of China on the Prevention and Control of Water Pollution, it is confirmed the project activity is in complicate with all laws and regulations in China.

Participation under other GHG programs

Projects registered (or seeking registration) under other GHG program(s): The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. A declaration from PP has been provided and checked by CTI.

Also, the assessment team checked the registries of CDM EB and Goldstandard through website <http://cdm.unfccc.int/> and <http://www.goldstandard.org/> to confirm the same.

Rejection by other GHG programs: The Project is not rejected by other GHG programs. The assessment team checked the registries of CDM EB, Goldstandard and Verra through website <http://cdm.unfccc.int/>, <http://www.goldstandard.org/> and <http://verra.org/> to confirm the same.

Other forms of credit and supply chain (Scope 3) emissions

Emissions trading programs and other binding limits: The assessment team confirms that the net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. The assessment team checked the REC Mechanism database of China and Chinese Emission Trading System (Chinese ETS) and found that the project activity is not accredited / registered under REC mechanism or Chinese ETS. Further, Declaration in effect of the same has been submitted by project proponent to the assessment team and found to be correct. Thus, the assessment team concluded that the project activity not involved on other Emissions trading programs and other binding limits.

Furthermore, as per "Notice on Strengthening Enterprise Greenhouse Gas Emission Report Management" /19/ issued by National Development and Reform Committee of P. R. China, China has a national emissions trading scheme only cover the high-emission industries, including thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year. By checking "Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022" /18/ issued by Ministry of Ecology and Environment of P.R.China, China has a national emissions trading scheme only covering thermal power generation industry. The project activity is not included in the mandatory emission control scheme and thus no emission cap was enforced for the project activity.

Other forms of environmental credit sought or received and eligible to be sought or received: The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

Renewable energy certificates are available for trading in the host country. However, the same is not availed by the project proponent. The undertaking regarding the same is submitted by PP which is acceptable to the assessment team. the assessment team also checked the REC website and found the declaration to be correct.

Issuance of public statement(s) to help prevent Scope 3 emissions double claiming: The main purpose of the project activity is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City, China. No public statement preventing potential risk of Scope 3 emissions double claiming has been made by PP or authorized representative currently. As per VCS Standard version 4.5, scope 3 emissions section will be effective for all project requests submitted on or after 1 March 2024.

Email notification of the potential risk of Scope 3 emissions double claiming: The main purpose of the project activity is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City, China. No public statement preventing potential risk of Scope 3 emissions double claiming has been made by PP or authorized representative currently. As per VCS Standard version 4.5, scope 3 emissions section will be effective for all project requests submitted on or after 1 March 2024.

Additional information relevant to the project

Leakage management for AFOLU projects: The project activity is not a AFOLU project, therefore, this section is not applicable.

Commercially sensitive information: No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.

Sustainable development contributions: The project activity would contribute sustainable development in the region in following aspects confirmed by site visit:

By checking EIA Report /16/, Employment Roster /66/, ER calculation sheet /2/, Solid Liquid Separation Record /45/, China's National Plan on Implementation of the 2030 Agenda for Sustainable Development, China's 14th Five-Year Plan and the 2035 Vision Outline /22/, and 17 SDGs defined by UNDP, interviewing stakeholders during site visit, the validation team confirmed that the project activity would contribute sustainable development in the region and China's Sustainable Development Goals (SDGs). The specific sustainable development goals that the project activity could achieve have been listed as follows:

- **SDG13** "Take urgent action to combat climate change and its impacts": The main purpose of the project activity is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City, China. This contributes to achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas". During this monitoring period, total GHG emission reduction was 88,210 tCO₂e and the cumulative Contributions Over Project Lifetime were 88,210 tCO₂e achieved emission reductions. The detailed ER calculation has been included in ER calculation sheet /2/ and verified by the assessment team to be correct.

- **SDG 8** "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all": the project activity will increase employment opportunities for the operation of the project activity. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value". By checking Employment Roster /66/, it is confirmed since operation of the project activity (project start date), 18 long term working opportunities are provided.

- **SDG 12** "Ensure sustainable consumption and production patterns": The main purpose of the project activity is to avoid methane emissions and produce organic solid fertilizers through aerobic composting solids in animal manure from swine farms in Leiyang City, China. By checking FSR /14/, it was verified the project activity could treat approximately 11,550 t Volatile Solids (on a dry matter weight basis) contained in the animal manure waste annually. The project activity will ensure sustainable consumption and production patterns, which contributes to one of China's actions for promoting sustainable developing: "Promote circular production methods, promote green design and clean production, improve waste materials' recycling network, and enhance renewable resource recycling

capacity. Promote the formulation and revision of relevant standards and specifications for the identification of solid and hazardous waste, promote waste classification, comprehensive utilization of industrial solid waste, and recycling and disposal of agricultural waste, and promote the high-quality development of the domestic renewable resource industry. Deepen the pilot reform and innovation of "zero-waste city" construction, and play a leading role in the demonstration". By checking Solid Liquid Separation Record /45/, it was verified, during this monitoring period, approximately 17,355 t Volatile Solids (on a dry matter weight basis) contained in the animal manure waste was treated and the cumulative Contributions Over Project Lifetime were 17,355 t Volatile Solids (on a dry matter weight basis) contained in the animal manure treated. The detailed Volatile Solids calculation has been included in ER calculation sheet /2/ and verified by the assessment team to be correct.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. The Project is not rejected by other GHG programs. A declaration from PP has been provided and checked by CTI.

Also, the assessment team checked the registries of CDM EB and Goldstandard through website <http://cdm.unfccc.int/> and <http://www.goldstandard.org/> to confirm the same.

Therefore, the assessment team confirmed that the joint PD & MR is accurate, complete, and provides an understanding of the nature of the project.

3.3 Safeguards

3.3.1 No Net Harm

The Environmental Impact Assessment (EIA) Report of the project activity was compiled by Chongqing Fengda Environmental Impact Assessment Co., LTD /16/ and approved by Leiyang City Ecological Environmental Bureau on 15/01/2021 /17/. The assessment team confirm all environmental impacts have been analyzed and no net harm was detected. Refer to section 3.3.3 of this report for detailed environmental impacts arising from the Project construction and operation.

Through interview with local stakeholders, it is confirmed by the assessment team the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 3.2 of this report above.

In conclusion, it is confirmed by the assessment team the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The assessment team checked the relevance of the dates during the validation site visit.

By checking the attendance list, the assessment team confirmed a local stakeholder consultation survey was conducted to collect comments from local residents through holding stakeholder meeting and distributing questionnaires in May 2019 by the project owner. Through checking questionnaires /30/, as well as interviewing with the project owner and local stakeholder, the assessment team confirmed that the questionnaire includes a brief summary of the project activity, including the project design and other relevant key information. In total 30 out of 30 questionnaires were returned with a 100% response rate.

By checking the questionnaires /30/, as well as interviewing with the project owner and local stakeholder, the assessment team confirm that the local stakeholder has no negative comments for the construction of the project activity. As confirmed with PP, stakeholder meetings will be held at periodic intervals to keep on-going communications with various stakeholders. Besides, local government agencies and competent authorities will conduct spot checks on the implementation of the project. Therefore, the assessment team was able to confirm that the stakeholder meeting was adequate and appropriate.

By checking the filled questionnaires from local stakeholders /30/, the assessment team confirmed that communications with local stakeholders was carried out in November 2022 by the project owner through distributing questionnaires for this monitoring period. By checking the questionnaires /36/, it was verified no negative comments received for the project during this monitoring period.

Besides, during site visit, it is confirmed a grievance book /73/ has been located at the office of the project site. No negative comments have been identified in the grievance book.

Furthermore, via interviewing with staff from local government during the site visit, it is confirmed during the implementation stage of this monitoring period, local authority has conducted spot checks and the project passed all the periodic spot checks by local government.

Therefore, the assessment team was able to confirm that PP has established ongoing communications with stakeholders as required by VCS Standard version 4.5.

3.3.3 Environmental Impact

An Environmental Impact Assessment has been conducted by the project participants. The assessment team has reviewed the documentation of the presented information, eg. EIA Report /16/, EIA Approval /17/ and Completion Acceptance Opinion /70/. The EIA Approval /17/ and Completion Acceptance Opinion /70/ confirms the correctness of the approach used by PPs. In conclusion, the PPs have followed the requirements of the host country with regards to addressing environmental impacts as below:

Air Pollution

By checking EIA Report /16/, as well as through site inspection and interviewing with stakeholders, the assessment team confirmed the collection and transportation of swine manure has been in an enclosed environment and the composting workshop is designed fully enclosed and was equipped with ventilation facilities. By checking EIA Approval /17/ and via on site investigation, it was verified the

odorous gases are purified through a biofilter tower to meet the secondary standard of the national standard "Emission Standard for Odor Pollutants" (GB14554-93) and afterwards discharged by the 15m exhaust cylinder; the exhaust gases from the crushing and screening processes of composting undergo dust collection and are treated by a bag filter to meet the secondary standard of the Comprehensive Emission Standard of Air Pollutants (GB16297-1996) can be satisfied and afterwards it is discharged through the 15m exhaust cylinder. Also, the project takes the solid fertilizer workshop as the boundary, and sets a health protection distance of 100m. Besides, by checking Equipment Purchase Agreement /24//25/, and via on site investigation, it was verified bag dust collector has been identified in operation at the project site.

Wastewater

By checking EIA Approval /17/ and via on site investigation, it was verified the wastewater of the project activity includes rainwater, surface cleaning wastewater, and production wastewater (wastewater from equipment and vehicle cleaning). The wastewater is collected and fed into the pre-treatment pool without being discharged into the external river system. The transportation of manure is carried out using specialized enclosed vehicles, and strict storage and transportation procedures are established to ensure the cleanliness of the ground inside and outside the project site. The solid fertilizer workshop is designated as a general anti-seepage area. Besides, via checking Operation Log /34/ and Equipment Purchase Agreement /24//25/, as well as via on site investigation, it was verified liquid fertilizer treatment equipment has been identified in operation at the project site, the separated liquid together with natural wastewater is treated as liquid fertilizer.

Noise

By checking EIA Approval /17/ and via on site investigation, as well as through site inspection and interviewing with staff from the project owner, the assessment team confirmed that the noise mainly includes operational sounds from equipment, such as grinding and cutting pumps, crushers, and screening machines, as well as transportation noise from vehicles. Measures such as reinforcement and vibration reduction, installation of soundproofing devices, sealing, and rational scheduling of production time should be implemented. The plant boundary noise of the project should meet the requirements of Class 2 standard of "Industrial Enterprise Plant Boundary Environmental Noise Emission Standard" (GB12348-2008). Besides, via site investigation and interview with the project owner, it is confirmed the noise has been kept in reasonable range during working time. Via checking online satellite image, it was verified the project was not located in residential area.

Solid Waste

By checking EIA Approval /17/, Operation Log /34/ and via on site investigation, as well as through site inspection and interviewing with staff from the project owner, the assessment team confirmed the dust collected by the bag filter in this project can be returned for production. Besides, via on site investigation, it was verified greening has been carried out on the project site in order to alleviate dust and odors.

In conclusion, after the above measures are performed, the negative impacts on environment will be minimized below the requirements of laws and regulations during the construction and operational period.

3.3.4 Public Comments

The assessment team noted that this project was open for public comment from 22/11/2023 to 22/12/2023. The detail was checked by the assessment team in the following web platform: <https://registry.verra.org/app/projectDetail/VCS/4794>.

During the period, no public comments were received.

3.3.5 AFOLU-Specific Safeguards

The project activity is not an AFOLU project.

For non-AFOLU projects, this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The assessment team checked that following methodology and tools are applicable for the project activity as below:

Title	Methane avoidance through separation of solids from wastewater or manure treatment systems, AMS-III.Y., Version 4.0
Reference	The project activity meets the eligibility criteria of small-scale project as: - the estimated annual average GHG emission reductions or removal per year is 57,307 tCO ₂ e which is less than 60,000 tonnes of CO ₂ e per year.
Methodology	Methane avoidance through separation of solids from wastewater or manure treatment systems, AMS-III.Y., Version 4.0
Sectoral scope(s)	13
Category	Approved Consolidated Methodology
Tools	Tools referred with above methodology and applicable for project activity are: - AMS-III.F: Avoidance of methane emissions through composting (Version 12.0); - AMS-I.D: Grid connected renewable electricity generation (Version 18.0); - Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 03.0); - Tool 13: Project and leakage emissions from composting (Version 02.0); - Tool 21: Demonstration of additionality of small-scale project activities (Version 13.1)

3.4.2 Applicability

The applied baseline methodology AMS-III.Y., Version 4.0 is justified as it has been demonstrated that the proposed project activity is:

Applicability criteria	Justification of the project situation
This methodology comprises technologies and measures that avoid or reduce methane production from anaerobic wastewater treatment systems and anaerobic manure management systems, through removal of (volatile) solids from the wastewater or manure slurry stream. The separated solids shall be further treated, used or disposed in a manner resulting in lower methane emissions.	By checking FSR /14/, List of livestock farms /67/ and through on-site interview, the assessment team confirmed that the project activity uses a solid-liquid separation system to separate solids in swine manure from existing swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. By checking FSR /14/ and through on-site inspection (photos have been taken by the assessment team) and interview with staff from swine farms, it was verified before the implementation of the project activity, animal manure was left to decay anaerobically in uncovered anaerobic lagoons in existing swine farms within the project boundary and the generated methane is emitted to the atmosphere. The project activity could treat approximately 52,500 tonnes of manure and could achieve average annual GHG emission reductions of 57,307 tCO₂e annually. Therefore, it is confirmed this criteria is applicable.
The project activity does not recover and combust biogas i.e. the baseline wastewater or manure treatment plant as well as the project system are not equipped with methane recovery system. Project activities which recover and combust biogas from manure management systems shall consider AMS-III.D or AMS-III.R. Project activities which recover and combust biogas from wastewater treatment systems shall consider AMS-III.H. Project activities that substitute anaerobic wastewater treatment systems with aerobic wastewater treatment system shall consider AMS-III.I.	By checking FSR /14/ and through on-site interview, the assessment team confirmed that the project activity uses a solid-liquid separation system to separate solids in swine manure from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. It is confirmed the project activity does not recover and combust biogas, i.e. the project system are not equipped with methane recovery system. Besides, via site inspection (photos have been taken by the assessment team) and interview with staff from local swine farms, it is confirmed all the swine farms in Leiyang City are not involved in methane recovery, i.e. the baseline manure treatment plant are not equipped with methane recovery system. Therefore, this criteria is applicable.
The technology for solids separation shall be one of the below or a combinations thereof so as to achieve a minimum dry matter content of	By checking FSR /14/, Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and through on-site interview and inspection,

separated solids larger than 20% a) A pre-separation phase of chemical treatment by mixing flocculants with the wastewater, adopted to improve the efficiency of the subsequent mechanical solid-liquid separation process; b) Mechanical solid/liquid separation technologies (e.g. stationary, vibrating or rotating screens, centrifuges, hydrocyclones, press systems/screws), operated in-line with the inflowing freshly generated wastewater or slurry manure stream so as to avoid stagnation; c) Thermal treatment technologies that evaporate water content from the waste stream, either releasing vapour to the atmosphere or condensing it into a liquid fraction (condensate). Examples include evaporation and spray drying technologies.	the assessment team confirmed that the technology for solids separation applied by the project activity belongs to (b) mechanical solid/liquid separation technologies. Besides, as per industry standard “General Technical Conditions for Solid-Liquid Separation of Livestock and Poultry Manure” (JB/T 13756-2019), “when the separator machine separates solid materials using the filter specified in the product manual under normal temperature and pressure, the overall performance of the machine should meet the following criteria: for separators used for flushing (bubbling) manure, the moisture content of the separated solid material should be $\leq 75\%$”. By checking Solid-liquid separation Record /59/, it was verified the moisture content of the separated manure after solid-liquid separation is around 60%, which further confirmed that the dry matter content of separated solids could remain higher than 20% throughout until its final use as fertilizers.
The dry matter content of separated solids shall remain higher than 20% throughout until its final disposal, destruction or use (e.g. spreading on the soil). The total time interval for the separation process until 20% dry matter content is achieved shall be less than 24 hours.	By checking Solid-liquid separation Records /59/, it was verified the dry matter content of separated solids shall remain higher than 20% throughout until its final use. By checking Operation Log /34/, site inspection and interviewing with PP, it was verified solids from the collected swine manure will be separated the day arrived at the project site, therefore, the assessment team confirmed the total time interval for the separation process until 20% dry matter content is achieved within 24 hours. Therefore, this criteria is applicable.
Separation of solids using gravity (settling tanks/basins, ponds, or geotextile containers/bags) is not included in this methodology.	By checking FSR /14/, Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and through on-site interview and inspection, the assessment team confirmed that the separation of solids using gravity (settling tanks/basins, ponds, or geotextile containers/bags) is not involved in the project activity. Therefore, this criteria is applicable.
In case of animal manure management systems the following conditions apply: (a) Animals shall be managed in confined conditions; (b) If organic bedding material is used in the animal barns or added to the manure stream,	By checking FSR /14/ and through on-site interview and inspection, the assessment team confirmed that: (a) By checking Livestock Manure Treatment Agreement /69/, it was verified the project activity treats swine manure from swine farms in Leiyang

separated solids shall not be reused in the barns; (c) If the baseline manure slurry was treated in an anaerobic lagoon or another liquid treatment system, the outflow liquid from the lagoon was recycled as flush water or used to irrigate fields; however it was not discharged into river/lake/sea. In the latter case, i.e. effluent discharge into river/lake/sea, the system is considered as a wastewater treatment system and not a manure management system; (d) A minimum interval of six months was observed between each removal of the solids accumulated in the lagoon.	City, where the animals are managed in confined conditions; (b) By checking FSR /14/ and through on-site interview, the assessment team confirmed that the project activity uses a solid-liquid separation system to separate solids for further producing organic solid fertilizers through aerobic composting, no separated solids will be reused in the barns; (c) By checking FSR /14/, through site inspection (photos have been taken by the assessment team) and on-site interview, the assessment team confirmed the baseline manure slurry was treated in an anaerobic lagoon, the outflow liquid from the lagoon was used to irrigate fields, not discharging into natural water body. (d) Via onsite interviewing with livestock farmers, it was verified the animal manure from swine farms in Leiyang City was left to decay anaerobically in uncovered anaerobic lagoons and was used for land application in nearby farms. By checking public information /75/, it was verified in order not to destroy the crops and seedlings in the nearby farms, the manure (including solids) would be accumulated in the lagoon for more than 6 months, therefore, the assessment team confirmed a minimum interval of six months was observed between each removal of the solids accumulated in the lagoon. In conclusion, this criteria is applicable.
In case of wastewater treatment systems the following conditions apply: (a) The baseline treatment systems do not include a fine solids separation process (i.e. grading smaller than 10 mm aperture, primary settlers, mechanical separation, etc.); (b) In case the baseline treatment system was an anaerobic lagoon or a liquid system, a minimum interval of 30 days was observed between each removal of the solids accumulated in the lagoon.	By checking FSR /14/ and through on-site interview, the assessment team confirmed no wastewater treatment systems were involved in the project activity or baseline uncovered anaerobic lagoons. Therefore, this criteria is not relevant.
This methodology is not applicable when the project treats solids removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device of the baseline animal manure management/wastewater treatment system.	Via onsite inspection and interview with PP, it was verified the project uses a solid-liquid separation system to separate solids from manure connected from the swine farms in Leiyang City, not removed from an existing lagoon, or sludge originated from settlers or from any other biologically active treatment device of the baseline animal manure management/wastewater treatment system.

	Therefore, this criteria is not relevant.
The separated solids shall be further treated and methane emissions resulting from further treatment, storage, use or disposal shall be considered. If the solids are combusted for thermal or heat generation, that component of the project activity can use a corresponding methodology under Type I. If the solids are mechanically/thermally treated to produce refuse-derived fuel (RDF) or stabilized biomass (SB) the relevant provisions in AMS-III.E shall be followed. If the solids are used as animal feeds (e.g. feed to cows, pigs), emissions from enteric fermentation and emissions from the manure may be neglected.	By checking FSR /14/, Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and through on-site interview, the assessment team confirmed that the project activity uses a solid-liquid separation system to separate solids in swine manure from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. Methane emissions resulting from aerobic composting of separated solid has been accounted as part of the project emissions. Therefore, it is confirmed this criteria is applicable.
The liquid fraction from the project solid separation system shall be treated either in the baseline treatment facility or in a treatment system with a lower methane conversion factor (MCF) than the baseline system.	By checking FSR /14/ and Equipment Purchase Agreement /24/, as well as onsite inspection and interviewing with PP, it was verified the liquid fraction from the project solid separation system is diverted to produce liquid fertilizers in a water management system that combines aerobic and low-leakage anaerobic treatments. By checking Table 10.17 (updated) (continued) from 2019 Refinement to the 2006 IPCC Guidelines, it was verified MCF for both aerobic treatment and anaerobic digesters are much lower than that of uncovered anaerobic lagoons. Therefore, it was verified the liquid fraction from the project solid separation system was treated in a treatment system with a lower MCF than the baseline uncovered anaerobic lagoons. Therefore, it is confirmed this criteria is applicable.
This methodology applies to situations where the baseline treatment systems have been operational at least three years before the start date of the project activity. New facilities (Greenfield projects) and project activities involving a change of equipment resulting in an efficiency improvement or capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the requirements in the “General Guidelines to SSC CDM methodologies” concerning these topics. In addition, the requirements for demonstration of the remaining lifetime of the equipment replaced as described in the general guidelines shall be followed.	By checking operation years for each livestock farm contained in List of livestock farms /67/ and via onsite interviewing with livestock farmers, it was verified the uncovered anaerobic lagoons in local swine farms have been operated at least 3 years before the start date of the project activity (28/03/2022), which was a very traditional and long-standing common practice among local swine farms. Furthermore, by checking Livestock Manure Treatment Agreement /69/, and based on a review of publicly available information provided by the National Enterprise Credit Information Publicity System (developed by an officially registered enterprise credit agency, Qichacha), it was verified all the entities operating all the involved swine farms have been operational for at least three

	years prior to the implementation date of the project (28/03/2022).
In case flocculants is used in the project activities, project emissions and leakage from use of flocculants should be taken into account.	By checking FSR /14/, via onsite inspection and interviewing with PP, it was verified no flocculants have been used in the project activity. Therefore, it is confirmed this criteria is not relevant.
Measures are limited to those that result in emission reductions of less than or equal to 60 Kt CO ₂ equivalent annually.	The project activity could treat approximately 52,500 tonnes of manure and could achieve average annual GHG emission reductions of 57,307 tCO ₂ e annually, less than 60,000 tCO ₂ e. Therefore, it is confirmed this criteria is applicable.

As per para 34 of AMS-III.Y version 04.0, in case the solids separated from manure or from wastewater are composted, the methane emissions from composting of solids separated by the project activity ($PE_{s,composting,y}$) is determined as per the relevant procedure in AMS-III.F. As per para 45 of AMS-III.Y version 04.0, in case of composting of separated solids, relevant parameters for calculating the project emissions from the composting of the separated solids shall be monitored according to the relevant procedure in AMS-III.F.

As per para 29 (b) of AMS-III.Y version 04.0, emissions from electricity and fossil fuel use by the solid separation technology as per the methods of AMS-I.D. As per para 35 of AMS-III.Y version 04.0, project activity emissions from electricity or fossil fuel consumption ($PE_{y,power}$) are determined as per the procedures described in AMS-I.D.

Therefore, latest version AMS-III.F version 12.0 and AMS-I.D version 18.0 applied by the project activity was verified to be in compliance with AMS-III.Y version 04.0.

Applicability conditions of "Tool 13: Project and leakage emissions from composting (version 02.0)"

Applicability criteria	Justification of the project situation
Typical applications of the tool include projects composting municipal solid wastes, agricultural wastes and digestate.	By checking FSR /14/, Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and through on-site interview, the assessment team confirmed that the project activity uses a solid-liquid separation system to separate solids in swine manure (agricultural wastes) from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. Therefore, this criteria is applicable.
The following sources of project emissions are accounted for in this tool: a. CH ₄ and N ₂ O emission from composting;	By checking FSR /14/, Equipment Purchase Agreement /24//25/, nameplate of equipment /20/ and through on-site interview, the assessment team confirmed that the project

b. CO₂ emissions from consumption of fossil fuels and electricity associated with composting; and c. CH₄ emissions from run-off wastewater associated with co-composting.	activity uses a solid-liquid separation system to separate solids in swine manure from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. The project activity accounted for a. CH₄ and N₂O emission from composting and b. CO₂ emissions from consumption of fossil fuels and electricity associated with composting. The project activity does not involve co-composting and thereby c. CH₄ emissions from run-off wastewater associated with co-composting is not applicable.
The following source of leakage emissions is accounted for in this tool: (a) CH₄ emissions from the anaerobic decay of the residual organic content of compost disposed of in a landfill or subjected to anaerobic storage.	As justified above, the project activity uses a solid-liquid separation system to separate solids in swine manure from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. During site inspection, it was verified the produced organic solid fertilizers are stored and kept in packed bags under aerobic conditions. Therefore, no anaerobic decay of the residual organic content of composting will be disposed in a landfill or subjected to anaerobic storage. Therefore, this criteria is not applicable.
Transport emissions are not accounted for in this tool because it is assumed that similar transportation activities would occur in the baseline.	Similar transportation activities are assumed to occur in the baseline. Therefore, this criteria is applicable.

Applicability conditions of "Tool 05: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (Version 03.0) "

Applicability criteria	Justification of the project situation
If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants	By checking FSR /14/ and Electricity Purchase Receipts /47/, as well as site interview, it is confirmed the project activity will use electricity from local power grid which connected to Central China Power Grid (CCPG). Therefore, scenario A is applicable.

are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	By checking FSR /14/ as well as site interview, it is confirmed the project activity only involves electricity consumption, no electricity will be generated and supplied. Therefore, this criteria is not applicable.
This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.	By checking FSR /14/ as well as site interview, it is confirmed no captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. Therefore, this criteria is not relevant.

Applicability conditions of "Tool 21: Demonstration of additionality of smallscale project activities (version 13.1)"

Applicability criteria	Justification of the project situation
This methodological tool provides a general framework for demonstrating and assessing additionality and is applicable to a wide range of project types.	The project activity using AMS-III.Y. (Version 4.0) meets the eligibility criteria of small scale project as the estimated annual average GHG emission reductions or removal per year is 57,307 tCO₂e which is less than 60,000 tonnes of CO₂e per year. As per para 18 of AMS-III.Y. (Version 4.0), PP shall apply the general guidelines for the SSC CDM methodologies and the Tool for demonstration of additionality of SSC project activities. Therefore, this criteria is applicable.

In validating the application of this methodological tool, Designated Operation Entities (DOEs) shall carefully assess and verify the reliability and creditability of all data, rationales, assumptions, justifications and documentation provided by project participants to support the demonstration of additionality. The elements checked during this assessment and the conclusions shall be documented transparently in the validation report.	In validating the application of this methodological tool, VVB has carefully assess and verify the reliability and creditability of all data, rationales, assumptions, justifications and documentation provided by PP to support the demonstration of additionality. The elements checked during this assessment and the conclusions have been documented transparently in the joint validation&verification report. Details refer to section 3.3.5 of this report. Therefore, this criteria is applicable.
The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	No new methodologies are proposed and no alternative methods are used to demonstrate additionality. Besides, as justified above, the aggregate size of the project activity is less than SSC thresholds (emission reductions of the project activity is less than 60ktCO₂e/y), the project activity is not comprised of any technologies from the positive list under tool 32, and the aggregate size of the project activity is larger than MSC thresholds (emission reductions of the project activity is greater than 20ktCO₂e/y). As per "Appendix. Provisions of small-scale and microscale tools for automatic additionality" of the tool "Demonstration of additionality of small-scale project activities", such project activity use regular additionality procedure. Therefore, this criteria is not relevant.
Project participants and coordinating/managing entities may also apply "TOOL19: Demonstration of additionality of microscale project activities" as applicable.	As justified above, the aggregate size of the project activity is larger than MSC thresholds (emission reductions of the project activity is greater than 20ktCO₂e/y). Therefore, this criteria is not relevant.

The assessment team confirmed that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology i.e. AMS-III.Y., Version 4.0 is applicable to the project activity. The annual average GHG emission reductions or removal per year of the project activity is 57,307 tCO₂e, less than 60,000 tonnes of CO₂e per year. Therefore, the project is applicable as per small scale project activities methodology AMS-III.Y., Version 4.0.

3.4.3 Project Boundary

As per AMS-III.Y., the project boundary is the physical, geographical site:

(a) Where the animal waste would have been collected, stored and treated and the methane emission would have occurred in the absence of the proposed project activity;

- (b) Where the wastewater treatment would have taken place and the methane emission would have occurred in the absence of the proposed project activity;
- (c) Where the treatment of animal waste or wastewater through solids separation takes place;
- (d) Where the storage, gainful use, destruction and/or land application of the separated solids takes place;
- (e) The itineraries between them, where the transportation of separated solids occur.

By checking FSR /14/ and through on-site interview, the assessment team confirmed no wastewater treatment systems were involved in the project activity or baseline uncovered anaerobic lagoons, all separated solids undergo immediate on-site composting without storage, gainful use or destruction. Therefore, the project boundary of the project activity was verified to be:

- (a) Uncovered anaerobic lagoons in swine farms in Leiyang City where the swine manure would have been collected, stored and treated and the methane emission would have occurred in the absence of the project activity;
- (b) The project site where swine manure was treated through solid separation;
- (c) The project site where the separated solids were composted;
- (d) The site where land application of the separated solids takes place;
- (e) The itineraries where the transportation of matters within the project boundary.

This is in compliance with AMS-III.Y. version 04.0.

The sources and GHG gases involved for the Project activity are as below.

Source	Gas	Included?	Justification/Explanation	
Project Baseline	Baseline emissions from the waste treatment process	CO ₂	No	Excluded for simplification. This emission source is assumed to be very small.
		CH ₄	Yes	As justified above, before the implementation of the project activity, animal manure was left to decay anaerobically in uncovered anaerobic lagoons within the project boundary and was used for land application in nearby farms, the generated methane is emitted to the atmosphere. By checking section 5.2 of AMS-III.Y., Version 4.0, it was verified this is the major emission source.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	No	Excluded for simplification. This emission source is assumed to be very small.
	Project emissions from storage, use, destruction or	CO ₂	Yes	By checking para 13 of Tool 13 “Project and leakage emissions from composting” version 02.0

Source	Gas	Included?	Justification/Explanation
disposal of solids separated			as mentioned in section 26 of AMS-III.F. version 12.0, it was verified this maybe a major emission source.
			By checking FSR /14/ and via site inspection, it was verified this project activity consumes electricity and fossil fuel. Therefore, it was verified this is a major emission source.
	CH ₄	Yes	By checking para 30 of AMS-III.Y., version 4.0 and para 13 of Tool 13 “Project and leakage emissions from composting” version 02.0 as mentioned in section 26 of AMS-III.F. version 12.0, it was verified the composting generates methane. Therefore, this is a major emission source.
	N ₂ O	Yes	By checking para 13 of Tool 13 “Project and leakage emissions from composting” version 02.0 as mentioned in section 26 of AMS-III.F. version 12.0, it was verified the composting generates N ₂ O. Therefore, this is a major emission source.
	Other	No	Excluded for simplification. This emission source is assumed to be very small.
Emissions from electricity and fossil fuel use by the solid separation technology	CO ₂	Yes	By checking para 35 of AMS-III.Y., version 4.0, it was verified this maybe a major emission source.
			By checking FSR /14/ and via on-site inspection, it was verified the solid separation technology consumes only electricity. However, this part of emissions have been included in “Project emissions from storage, use, destruction or disposal of solids separated”. The detailed justification has been included in section 3.4.6 of this report.
	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Other	No	Excluded for simplification. This emission source is assumed to be very small.
Emissions from the combustion of flocculants used in the pre-separation phase	CO ₂	No	As per para 36 of AMS-III.Y., version 4.0, in case of combustion of separated solids, the CO ₂ emissions from the combustion of the non-biomass (i.e. fossil) carbon content of the flocculants shall be considered. By checking FSR /14/ and Operation Log /34/, as well as via onsite inspection, it was verified the project activity does not involve
	CH ₄	No	
	N ₂ O	No	
	Other	No	

Source	Gas	Included?	Justification/Explanation
Incremental CO ₂ emissions due to increased transportation			combustion of separated solids, therefore, these emission sources are not relevant.
	CO ₂	Yes	By checking FSR /14/ and via onsite investigation, it is confirmed this can be the major emission source.
	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Other	No	Excluded for simplification. This emission source is assumed to be very small.

3.4.4 Baseline Scenario

As per AMS-III.Y., the baseline scenario is the situation where the solids separated from manure stream or from the wastewater would be treated in the wastewater treatment or manure management system within the project boundary, without methane recovery, and methane is emitted to the atmosphere.

As per CDM Validation and Verification Standard for project activities version 03.0, "where the baseline scenario is not prescribed in the approved methodology, the DOE shall assess the list of identified credible alternatives to the project activity in the VCS PD selected to determine the most realistic baseline scenario." As the selected small scale methodology clearly mention the baseline scenario and the same has been opted in this project, therefore, no further analysis on baseline is required.

The assessment team confirms that the joint PD & MR conforms to the guidances, i.e. CDM Validation and Verification Standard for project activities version 03.0 and VCS standard Version 4.5.

Therefore, the assessment team confirms that the baseline scenario of the project activity is:

By checking FSR /14/, through site inspection (photos of uncovered anaerobic lagoon in swine farms where swines were collected by the project activity have been taken by the assessment team) and on-site interview, it was verified swine manure from swine barns in swine farms in Leiyang City is left to uncontrolled anaerobic decay in uncovered anaerobic lagoons on each farm within the project boundary. Via onsite interviewing with livestock farmers, it was verified after accumulating in the uncovered anaerobic lagoons for at least 6 months, the swine manure was used for land application in nearby farms. During this process, the generated methane from uncovered anaerobic lagoon is emitted to the atmosphere without any recovery.

As per para 3.14.1 of VCS Standard version 4.5, regulatory surplus means that project activities shall not be mandated by any law, statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework..

By checking laws and regulations public available, i.e. Environmental Protection Law of People's Republic of China, Environmental Impact Assessment Law of the People's Republic of China, Animal Husbandry Law of the People's Republic of China, Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution, Law of the People's Republic of China on the Prevention and Control of Air Pollution, Law of the People's Republic of China on the Prevention and Control of Water Pollution, it is confirmed the project activity is in compliance with all laws and regulations in China, it was verified no laws or regulations in China mandated using an aerobic animal manure composting system associated with solid separation techniques to avoid methane emissions through the separation of manure solids. Therefore, it was concluded that the project activity is regulatory surplus.

3.4.5 Additionality

As per AMS-III.Y, para 17, Project participants shall take into account the general guidance to the small-scale (SSC) clean development mechanism (CDM) methodologies and the Tool for demonstration of additionality of SSC project activities available at <http://cdm.unfccc.int/Reference/index.html>. As justified above, the aggregate size of the project activity is less than SSC thresholds (emission reductions of the project activity is less than 60ktCO₂e/y), the project activity is not comprised of any technologies from the positive list under tool 32, and the aggregate size of the project activity is larger than MSC thresholds (emission reductions of the project activity is greater than 20ktCO₂e/y). As per "Appendix. Provisions of small-scale and microscale tools for automatic additionality" of Tool 21, such project activity use regular additionality procedure. Therefore, Tool "Demonstration of additionality of small-scale project activities" (Version 13.1) is used to analyze the additionality of the project activity.

Based on Tool 21, project participants shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- (a) Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions;
- (b) Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- (c) Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- (d) Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

PP selected option (a), i.e. investment barrier, to demonstrate the additionality of the project activity. By checking FSR /14/ of the project activity, it is confirmed the project activity is not financially attractive

and will face investment barrier. However, a financially more viable alternative to the project activity (baseline scenario) would have led to higher emissions.

The parameters used in the financial calculations have been validated based on a review of the sources presented in the PD, including: FSR /14/ of the project activity, Table of financial benchmark rate of return of construction projects /26/ issued by NDRC and the Ministry of Construction on 15/03/2013 etc. The FSR /14/ of the project activity was completed by the project owner based on national regulations and has been approved by local government /15/. The finalization of initial version FSR /14/ (April 2019) and making investment decision /63/ at the same time, i.e. April 2019, therefore it's unlikely that the input values have significantly changed. Additionally, confirmation with the following documents and/or sources demonstrates that the parameters are plausible and acceptable under the project situation.

Choice of approach

As justified above, the baseline scenario of the project activity is "Swine manure from swine barns in swine farms in Leiyang City is left to uncontrolled anaerobic decay in uncovered anaerobic lagoons on each farm within the project boundary. After accumulating in the uncovered anaerobic lagoons for at least 6 months, the swine manure was used for land application in nearby farms. During this process, the generated methane from uncovered anaerobic lagoon is emitted to the atmosphere without any recovery", which is the continuation of current business-as-usual scenario and does not involve an investment from the project participants and has no barrier, therefore no investment analysis is required for this scenario.

As the project generates financial and economic benefits other than VCS related income through the sales of composting products. A Benchmark Analysis was chosen and conducted for investment analysis of the implementation of project activity without VCS revenue and the financial indicator - IRR, was chosen, which is verified to be reasonable.

Benchmark Selection

The project activity produces composting products, i.e. organic solid fertilizers. The IRR benchmark (10%) used for the financial analysis in the FSR /14/ of the project activity was obtained from the Table of financial benchmark rate of return of construction projects from "Notice on Update of Financial Benchmarks from Economic Evaluation for Construction Project: Methods and Parameters (Version 03)" issued by NDRC and the Ministry of Construction /27/. The benchmark IRR of project equity (post-tax) for fertilizer industry in China is 10%. As project activity produces composting products, the benchmark IRR of fertilizer industry in China can be applied to this project.

Thus, the assessment team was able to confirm the suitability of this benchmark.

Crosscheck of input parameters

The assessment team has verified and crosschecked all the input values used for the IRR calculations in the PD submitted for registration and issuance. The input parameters used in the financial analysis are sourced from FSR /14/ of the project activity.

FSR /14/ of the project activity is compiled by financial department of project owner based on national regulations, which can be regarded as a reliable and trustworthy source of information for financial analysis.

The assessment team compared the values stated in the PD with values determined in the FSR /14/ of the project activity and was able to confirm that the input values are applied correctly in the PD.

Calculation and comparison of financial indicators

Parameter name	Unit	Project data	Data source
Total Static Investment	CNY	28,000,000	FSR /14/
Annual O&M cost	CNY	8,362,277	FSR /14/
Annual organic solid fertilizers production	t	30,000	FSR /14/
Organic solid fertilizers production price	RMB	380	FSR /14/
Organic solid fertilizers Product VAT	/	9%	Announcement on deepening VAT reform related policies /35/
VAT tax refund	/	100%	Notice on VAT exemption of organic fertilizer products /37/
Income tax	/	25%	Income Tax Law of the People's Republic of China on Enterprises /41/
Rate of additional tax on city building	/	1%	Tax Law of the People's Republic of China on Urban Maintenance and Construction /38/
Rate of additional tax on education	/	5%	Decision of The State Council on amending the Interim Provisions on levying Additional Education Fees /39/
IRR	/	5.42%	Calculated

The accuracy and suitability of the input values for the investment analysis were crosschecked as below:

Total Static Investment: 28,000,000 CNY

Fixed assets: 27,478,991 CNY

By checking FSR /14/, it is confirmed total Static Investment is 28,000,000 CNY. By checking Settlement statement for completion of five sections of the project activity /51/-/55/, the assessment team confirmed that the actual investment cost of the project activity was 27,986,200 CNY. When applying actual static investment in IRR calculation, the IRR will be 5.43%, still much lower than the benchmark. still much lower than the benchmark. Therefore, the static total investment estimated in the FSR /14/ was accepted by the assessment team.

By checking FSR /14/, it is confirmed the fixed assets is 27,478,991 CNY and the equipment cost is 6,310,000 CNY. By checking Announcement on deepening VAT reform related policies /35/, it is confirmed the VAT of equipment is 9%. Considering 9% VAT rate of equipment, the deductible VAT cost of equipment is calculated to be $6,310,000 / (100\% + 9\%) * 9\% = 521,009$ CNY. Therefore, the fixed

assets is calculated to be total static investment - deductible VAT cost = 28,000,000 - 521,009 = 27,478,991 CNY.

O&M cost: 8,362,277 CNY

The O&M cost is consistent of annual maintenance cost, annual staff salary and welfare, annual insurance, Annual material fee and Annual other cost. Each component has been verified by the assessment team.

a) Annual maintenance cost: 549,580 CNY

The annual maintenance cost was estimated as 549,580 CNY in IRR calculation, which accounts for 2% of fixed asset investment. The assessment team did not find suitable guidelines for maintenance cost of the specific project activity. The rate 2% of annual maintenance cost is considered by the technical expertise in the assessment team as conservative because as per the publication issued by China Planning Press - Financial Evaluation Practice and Analysis of Difficult Problems-Feasibility study and evaluation of Bank loan projects /40/, such rate is 3%~5%.

b) Annual staff salary and welfare: 1,944,000 CNY

By checking FSR /14/, it is confirmed annual staff salary and welfare was estimated to be 18 persons*72,000 CNY/person year*(1+50%) (1,944,000 CNY in total). According to 2019 Hunan Provincial Statistic Yearbook /68/, the average wage of employees in Henan Province of year 2018 is 73,300 CNY/year, which is higher than the designed value. The number is appropriate considering operators and technicians should be available for the project activity. By checking Employment Roster /66/, it is confirmed 18 employees have been employed due to the implementation of the project activity, which is more that estimated. Therefore it's considered that the wage and welfare is reasonable.

c) Annual insurance: 68,697 CNY

The annual insurance was estimated as 68,697 CNY in FSR /14/, which accounts for 0.25% of the fixed assets. The assessment team did not find suitable guidelines for equipment insurance of the specific project activity. The rate 0.25% of fixed assets is considered by the technical expertise in the assessment team as conservative because the value was among the reasonable range (0.3%~0.6%) mentioned in Compilation method of municipal engineering investment estimation issued by Ministry of Construction of China /42/ and public available at internet.

d) Annual material fee: 4,000,000 CNY

The annual material fee was estimated as 4,000,000 CNY in FSR /14/, which is calculated as: unit material fee per tonne animal manure * the amount of animal manure collected per year = 40 CNY/tonne * 100,000 tonne = 4,000,000 CNY. By checking Manure Treatment Agreement /69/, it is confirmed the actual unit material fee per tonne animal manure collected is 40 CNY/tonne. The project capacity, i.e. the amount of separated solids and swine manure collected is fixed by local government /15/, and the ratio between separated solids and collected swine manure is relatively fixed. By

checking Statement about manure collected /77/ and Statement about Raw Material of Organic Fertilizers /74/, it was verified the amount a manure collected actually (from March 2022 to February 2023) amounted to 103,559 tons, higher than that estimated. Therefore, it is concluded that the annual material fee is conservative and reasonable.

e) Annual other cost: 1,800,000 CNY

The annual other cost was estimated to be 1,800,000 CNY in FSR /14/. which is calculated as: unit other cost per tonne organic fertilizer * the amount of organic fertilizer produced per year = 60 CNY/tonne * 30,000 tonne = 1,800,000 CNY. The project capacity, i.e. the amount of separated solids and swine manure collected, is fixed by local government /15/ and thereby the organic solid fertilizers produced is also fixed. The assessment team did not find suitable guidelines for unit other cost per tonne organic fertilizer of the specific project activity. By checking Statement about O&M cost /62/, it was verified the actual unit other cost per tonne organic fertilizer from March 2022 to February 2023 was 62 CNY/tonne, higher than that estimated in the FSR. Therefore, the annual material fee estimated is the FSR is conservative.

In general, each component of the O&M cost is crosschecked with plausible evidences.

Organic solid fertilizers revenue: 11,400,000 CNY

By checking FSR /14/, it is confirmed the organic solid fertilizers revenue is calculated via equation "organic solid fertilizers produced annually * the unit organic solid fertilizers production price". The organic solid fertilizers produced annually and the unit organic solid fertilizers production price were estimated to be 30,000 t and 380 CNY/t. By checking Organic Solid Fertilizers Sales Contract /61/, the assessment team confirmed that the actual organic solid fertilizers production price of the project activity was 400 CNY/t. When applying this unit price in IRR calculation, the IRR will be 7.85%, still much lower than the benchmark. Besides, the project capacity, i.e. the amount of separated solids and swine manure collected, is fixed by local government /15/ and thereby the organic solid fertilizers produced is also fixed. Besides, by checking Organic Fertilizer Statistics /72/, the assessment team confirmed the organic fertilizer production from March 2022 to February 2023 is 26,925 t, lower than that estimated in the FSR /14/, i.e. 30,000t/year. Therefore, it's confirmed that the organic solid fertilizers revenue in FSR /14/ is accepted by the assessment team.

Taxes: see below

The taxes applied in the financial assessment are listed as:

Items	Value
Value Added Tax (VAT) for composting products, i.e. organic fertilizers	9%
Value Added Tax (VAT) refund	100%
Urban maintenance and construction tax	1%
Surtax for education	5%
Income tax	25%

Value Added Tax (VAT) for composting products is determined as 9% in the FSR /14/ which is consistent with that in "Announcement on deepening VAT reform related policies" issued on 20/03/2019 /35/. Value Added Tax (VAT) refund for organic solid fertilizers is determined as 100% in the FSR /14/ which is consistent with that in "Notice on VAT exemption of organic fertilizer products (Caishui (2008) No. 56)" issued on 29/04/2008 /37/.

By cross checking with the national taxation regulations, sectoral regulations and other evidences, i.e. the Tax Law of the People's Republic of China on Urban Maintenance and Construction /38/ and Decision of The State Council on amending the Interim Provisions on levying Additional Education Fees /39/, the assessment team was able to confirm that the Urban maintenance and construction tax and Surtax for education in the financial assessment are all in line with national regulations.

An income tax of 25% is consistent with the Income Tax Law of the People's Republic of China on Enterprises /41/ issued by the Chairman of the People's Republic of China on 16/03/2007.

Depreciation period and residual rate: 15 years and 5%

As per FSR /14/, the residual rate is determined as 5% of the fixed assets, and the depreciation period is 15 years. As per Article 31 of the "Rules for the Provisional Regulations on Enterprise Income Tax of the People's Republic of China" /43/, the ratio of residual value shall be within 5% of the original value and shall be determined by the enterprise. Therefore, the assessment team was able to confirm that a rate of 5% is reasonable.

According to the "Enforcement Regulation of Income Tax Law of the People's Republic of China for Enterprises" /41/ issued by State Council on 06/12/2007, it stipulates that the depreciation period of production equipments should be at least 10 years. Thereby the 15 years of depreciation periods of the project activity is appropriate.

The depreciation rate of 6.33% is determined based on a depreciation period of 15 years and a residual rate of 5%.

Lifetime: 15 years

The lifetime of the project activity was estimated to be 15 years in the FSR /14/. By checking public information /26/, it is confirmed the life time of the equipment is 15 years. By checking FSR /14/, the 15-year operational period was determined primarily based on factors such as the project lifecycle, technological feasibility, and market demand, reflecting considerations for investment returns. To maintain consistency with the project lifecycle, the cutoff date for emission reduction calculations is set to the end of the second CP of the project. Therefore, the assessment team was able to confirm that the lifetime estimated in FSR /14/ is reasonable.

In conclusion, based on the information validated, the assessment team confirms that all the input values for investment analysis are in reasonable ranges. In addition, the assessment team was able to confirm that the input parameters used in the financial analysis are reasonable and adequately represent the economic situation of the project. Therefore, the assessment team confirms that the assumptions in investment analysis are appropriate and the financial calculations are correct.

Sensitivity analysis

2 variables that constitute more than 20% of total project costs, i.e. total static investment and annual O&M cost, and 2 variables that will impact total project revenues, i.e. annual organic solid fertilizers production and organic solid fertilizers price, are considered in the sensitivity analysis. As validated above, no other variables would constitute more than 20% of either total project costs or total project revenues. A departure variations of a range of +10% and -10% is applied which is consistent with the applicable CDM and VCS requirements.

As a result of sensitivity analysis, within the reasonable variation ($\pm 10\%$) of above variables, the IRR of project without VCS revenues is still lower than the benchmark.

Meanwhile, the assessment team assessed the possibility of the extreme points of the 4 variables to reach the benchmark 10%.

Total static Investment: if the total static investment decreased by 24.98%, the IRR would reach 10%. However, by checking Settlement statement for completion of five sections of the project activity /51/-/55/, the assessment team confirmed that the actual investment cost of the project activity was 27,986,200 CNY. When applying actual static investment in IRR calculation, the IRR will be 5.43%, still much lower than the benchmark. Therefore, it's unlikely that the static total investment decreased by 24.98%.

Annual O&M cost: if the annual O&M cost is decreased by 13.63%, the IRR would reach 10%. However, considering the increasing price index, the assessment team validated that the fluctuation of the variable will not make the IRR reach the benchmark. By checking Statement about O&M cost /62/, it was verified the actual O&M cost from March 2022 to February 2023 was 8,352,173 CNY, slightly less than that estimated in the FSR (i.e. 8,362,277 CNY), when applying actual O&M cost in IRR calculation, the IRR will be 5.47%, still much lower than the benchmark..

Annual organic solid fertilizer production: if annual organic solid fertilizer production is increased by 10.04%, the IRR would reach 10%. However, as the project capacity, i.e. the amount of swine manure treated, is fixed by local government /15/, and thereby the organic solid fertilizer produced is also fixed. Therefore, it is unlikely to increase the amount of products by 10.04%. By checking Organic Fertilizer Statistics /72/, the assessment team confirmed the organic fertilizer production from March 2022 to February 2023 is 26,925 t, lower than that estimated in the FSR /14/, i.e. 30,000 t/year. Therefore, the assessment team validated that the fluctuation of annual organic solid fertilizer production will not make the IRR reach the benchmark.

Organic solid fertilizer price: if organic solid fertilizers price is increased by 10.04%, the IRR would reach 10%. However, by checking Organic Solid Fertilizer Sales Contract /61/, the assessment team confirmed that the organic solid fertilizer production price of the project activity was 400 CNY/t. When applying this unit price in IRR calculation, the IRR will be 7.85%, still much lower than the benchmark. Therefore, the assessment team validated that the fluctuation of unit price of organic solid fertilizers will not make the IRR reach the benchmark.

It is assessment team's conclusion that sensitivity analysis used is appropriate and reasonable. Alternative (a), which is the implementation of project activity without VCS, is not financially attractive to the project owner.

3.4.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline emissions

As per para 23 of AMS-III.Y., in the case of manure stream, where the barns do not use any organic bedding materials, baseline emissions shall be calculated based on the total mass of the volatile solids separated as below:

$$BE_y = (B_{o,w,y} \times M_{ss,y} \times VS_{ss,y} \times UF_b \times GWP_{CH_4} \times D_{CH_4}/1000) \times \sum_i (MS_{Bl,i} \times MCF_{b,i}) \quad \text{Equation (1)}$$

Where,

- BE_y = Baseline emissions in year y (tCO₂e)
- $B_{o,w,y}$ = Weighted methane-producing potential of the volatile solids separated by the project in year y (m³ CH₄ per kg of VS). The project only involves manure from swine.
- $M_{ss,y}$ = Mass (dry matter basis) of total separated solids in year y (kg)
- $VS_{ss,y}$ = Volatile solids content of the separated solids in year y on a dry matter basis (kg/kg)
- UF_b = Model correction factor to account for model uncertainties (0.94)¹
- GWP_{CH_4} = Global Warming Potential of methane applicable to the crediting period (tCO₂e/tCH₄)
- D_{CH_4} = Conversion factor of m³ CH₄ to kilograms (0.67 kg per m³ at 20°C and 1 atm pressure)
- i = Index for baseline anaerobic manure management system
- $MS_{Bl,i}$ = Fraction of manure handled in the baseline anaerobic manure management system i (fraction, mass basis), based on historical information from previous year(s) before project. If historical information is not available, the capacity (t/day) of each baseline management system shall be verified before the project start. During the project, it would be assumed that the management systems with lower MCFs would be used up to their full capacity, and only thereafter the systems with larger MCF would be used. The value of $MS_{Bl,i}$ is "1.0" if only one type of treatment system was used in the baseline

¹ Reference: FCCC/SBSTA/2003/10/Add.2, page 25

$MCF_{b,i}$ = Methane conversion factor for the baseline anaerobic manure management system i as per IPCC Guidelines for National Greenhouse Gas Inventories

By checking FSR /14/, via on-site inspection (photos have been taken by the assessment team), as well as interviewing livestock farmers, it was verified the swine farms uses cement floor or manure leakage board, no any organic bedding materials were used in the barns for manure stream treated by the project activity. Therefore, the assessment team confirmed equation above is applicable for this project activity.

The parameters used to calculate $BE_{ex-ante}$ is determined and justified as below:

Parameter	Value	Data sources
$B_{o,w,y}$ ($B_{o,swine,y}$)	0.45 m ³ CH ₄ /kg-VS	By checking list of Livestock Farms /67/, Business Lisences of Livestock Farms /71/ and Statement about Raw Material of Organic Fertilizers /74/ issued by the project owner, it was verified the project activity collects only swine manure from about 16 swine farms in Leiyang City. By checking FSR /14/, list of Livestock Farms /67/, Business Lisences of Livestock Farms /71/ and through site inspection and interview, it was verified the swine farms are 100 percent market oriented with high level of capital input and high level of overall herd (flock) performance. Feed is purchased from local markets, swines are improved through breeding practices for commercial production and thereby the swine farms are with "High Productivity Systems". Therefore, IPCC default values from Table 10.16A of Volume 4 Chapter 10 of 2019 Refinement to the 2006 IPCC Guidelines for "swine" in "Other Region" with "High Productivity Systems" was selected and verified to be reasonable .
$M_{ss,y}$	21,000,000 kg	By checking FSR /14/, it was verified the mass (dry matter basis) of total separated solids was estimated to be 21,000,000 kg annually.
$VS_{ss,y}$	0.55 kg/kg	By checking FSR /14/, it was verified volatile solids content of the separated solids on a dry matter basis was estimated to be 0.55 kg/kg.
GWP_{CH_4}	28 tCO ₂ e/tCH ₄	IPCC Fifth Assessment Report (AR5)
D_{CH_4}	0.67 kg/m ³	As per AMS-III.Y, Conversion factor of m ³ CH ₄ to kilograms is 0.67 kg per m ³ at 20°C and 1 atm pressure).
UF_b	0.94	As per AMS-III.Y, model correction factor to account for model uncertainties is 0.94, which is sourced from FCCC/SBSTA/2003/10/Add.2, page 25.
$MS_{BI,i}$	100%	By checking FSR /14/ and via on-site inspection (photos of baseline treatment system have been taken by the assessment team) and interviewing livestock farmers, it was verified only one

		type of treatment system, i.e. uncovered anaerobic lagoon, was used in the baseline scenario. Therefore, as per AMS-III.Y, the value of $MS_{BI,i}$ is "1.0".
$MCF_{b,i}$	73%	By checking FSR /14/, it was verified there was only one type of treatment system existed prior to the project implementation. By checking public information /21/, it is confirmed the climate zone of project site belongs to Warm Temp. Moist. Therefore, default value of Uncovered anaerobic lagoon in climate zone of "Warm Temp. Moist", i.e. 73%, sourced from Table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 was applied.
BE_{CH4}	66,907 tCO ₂ e	Calculated via equation (1).

Project emissions

As per para 29 of AMS-III.Y. (Version 4.0), project activity emissions consist of:

- (a) Any methane emissions from storage, use, disposal or destruction of solids separated;
- (b) Emissions from electricity and fossil fuel use by the solid separation technology as per the methods of AMS-I.D;
- (b) Emissions from the combustion of flocculants used in the pre-separation phase;
- (c) Incremental CO₂ emissions due to increased transportation ($PE_{y,transp}$):
 - (i) Transportation of solids to sites where it will be treated further or gainfully used (within the project boundary);
 - (ii) Transportation of solids from and to treatment facilities to storage sites (within the project boundary);
 - (iii) Transportation of solids to disposal site.

$$PE_y = PE_{y,ss} + PE_{y,power} + PE_{y,floc,combustion} + PE_{y,transp} \quad \text{Equation (2)}$$

Where:

- $PE_{y,ss}$ = Project emissions from storage, use, destruction or disposal of solids separated in year y (tCO₂e).
- $PE_{y,power}$ = Project emissions from energy use for operating the separation device in year y (tCO₂e), calculated as per AMS-I.D methods
- $PE_{y,floc,combustion}$ = Project emissions from combustion of flocculants in year y (tCO₂e)
- $PE_{y,transp}$ = Project emissions for incremental transportation of matters in the project scenario, beyond the emissions for transportation of solids in the baseline scenario (tCO₂e)

✓ **Project emissions from storage, use, destruction or disposal of solids separated in year y ($PE_{y,ss}$)**

By checking FSR /14/, interviewing PP during site visit and via onsite inspection, it was verified no separated solid is stored, or destructed or disposed on site, all the solids are used immediately for composting after separation.

As per para 34 of AMS-III.Y. (Version 4.0), in case the solids separated from manure or from wastewater are composted, the methane emissions from composting of solids separated by the project activity ($PE_{s,composting,y}$) is determined as per the relevant procedure in AMS-III.F. Therefore,

$$PE_{y,ss} = PE_{s,composting,y} \quad \text{Equation (3)}$$

As per para 26 of AMS-III.F Version 12.0, project emissions from composting process shall be determined as per the latest version of the methodological tool “Project and leakage emissions from composting” is equivalent to parameter in the tool. As per para 13 of version 02.0 of this tool.

$$PE_{s,composting,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{N_2O,y} + PE_{RO,y} \quad \text{Equation (4)}$$

Where:

$PE_{s,composting,y}$	=	Project emissions associated with composting in the year y (tCO ₂ e)
$PE_{EC,y}$	=	Project emissions from electricity consumption associated with composting in year y (tCO ₂ /yr)
$PE_{FC,y}$	=	Project emissions from fossil fuel consumption associated with composting in year y (tCO ₂ /yr)
$PE_{CH_4,y}$	=	Project emissions of methane from the composting process in year y (tCO ₂ e/yr)
$PE_{N_2O,y}$	=	Project emissions of nitrous oxide from the composting process in year y (tCO ₂ e/yr)
$PE_{RO,y}$	=	Project emissions of methane from run-off wastewater associated with co-composting in year y (tCO ₂ e/yr)

Determination of the quantity of waste composted (Q_y)

As per para 14 of tool "Project and leakage emissions from composting", the quantity of waste composted is a parameter required in the determination of emissions associated with each source of project emissions. There are two options to determine the quantity of waste composted in year y (Q_y).

PP selected "Option 1: Procedure using a weighing device", i.e. monitor the weight of waste delivered to the composting installation using an on-site scale or any other applicable and calibrated weighing device (e.g. belt-scales), which is verified by the assessment team to be reasonable.

Determination of project emissions from electricity consumption ($PE_{EC,y}$)

As per para 17 of Tool 13 "Project and leakage emissions from composting" (version 02.0), where the composting activity involves electricity consumption from the grid or from a fossil fuel fired on-site power plant, $PE_{EC,y}$ shall be calculated using the latest approved version of the methodological tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation", where the project emission source j referred to in the tool is composting.

As per para 18 of AMS-III.Y. (Version 4.0), when applying this tool, if monitored data for electricity consumption is not available, then electricity consumption from composting ($EC_{PJ,comp,y}$) may be determined based on a default value for the specific quantity of electricity consumed per tonne of waste composted ($SEC_{comp,default}$), according to equation below. Note that the methodological tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" also provides options to calculate emission based on non-monitored parameters, including a default emission factor for the emissions per MWh of electricity consumed and an option to estimate electricity consumption based on the rated capacity of the captive power plant (if applicable).

As per Tool 05: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (Version 03.0), in the generic approach, project, baseline and/or leakage emissions from consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses. Therefore, the project emissions from electricity consumption was calculated via equation below:

$$PE_{EC,y} = EC_{PJ,composting,y} \times EF_{EF,composting,y} \times (1 + TDL_{composting,y}) \quad \text{Equation (5)}$$

Where:

$PE_{EC,y}$	=	Project emissions from electricity consumption in year y (t CO_2 / yr)
$EC_{PJ,composting,y}$	=	Quantity of electricity consumed by composting in year y (MWh/yr)
$EF_{EF,compost,y}$	=	Emission factor for electricity generation for composting in year y (t CO_2 /MWh). Default value of 1.3 t CO_2 /MWh suggested by Tool 05 shall apply
$TDL_{composting,y}$	=	Average technical transmission and distribution losses for providing electricity to composting in year y . Default value of 20% suggested by Tool 05 shall apply

During site inspection and interviewing with PP, it was verified the monitored data for quantity of electricity consumed by composting ($EC_{PJ,compost,y}$) is not available since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively

measured and recorded. Therefore, electricity consumption from the project activity is used as $EC_{PJ,compost,y}$ in calculation of $PE_{EC,y}$, which is conservative.

Determination of project emissions from fossil fuel consumption ($PE_{FC,y}$)

Where the composting activity involves fossil fuel consumption, PP may choose between the following two options to calculate $PE_{FC,y}$:

Option 1: Procedure using monitored data

$PE_{FC,y}$ shall be calculated using the latest approved version of the "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion", where the project emission source j referred to in the tool is composting.

Option 2: Procedure using a default value

Project emissions from fossil fuel consumption associated with composting are calculated as follows:

$$PE_{FC,y} = Q_y \times EF_{FC,default} \quad \text{Equation (6)}$$

Where

- $PE_{FC,y}$ = Project emissions from fossil fuel consumption associated with composting in year y (t CO_2 / yr)
- Q_y = Quantity of waste composted in year y (t/yr)
- $EF_{FC,default}$ = Default emission factor for fossil fuels consumed by the composting activity per tonne of waste (t CO_2 /t)

PP selected **option 2** in determine project emissions from fossil fuel consumption associated with composting. As per the "Data / Parameter table 5" of Tool "Project and leakage emissions from composting" (Version 02.0), $EF_{FC,default}$ is 0.0207 t CO_2 /t.

Determination of project emissions of methane ($PE_{CH_4,y}$)

As per paragraph 22 of Tool "Project and leakage emissions from composting" (Version 02.0), project emissions of methane from composting ($PE_{CH_4,y}$) are determined as follows:

$$PE_{CH_4,y} = Q_y \times EF_{CH_4,y} \times GWP_{CH_4} \quad \text{Equation (7)}$$

Where

- $PE_{CH_4,y}$ = Project emissions of methane from the composting process in year y (t CO_2e / yr)
- Q_y = Quantity of waste composted in year y (t/yr)
- $EF_{CH_4,y}$ = Emission factor of methane per tonne of waste composted valid for year y (t CH_4 / t)
- GWP_{CH_4} = Global Warming Potential of CH_4 (t CO_2e / t CH_4)

There are two options which project participants may choose for determining $EF_{CH_4,y}$:

Option 1: Procedure using monitored data

$EF_{CH_4,y}$ is determined based on measurements of the methane emissions during a composting cycle ($ECC_{CH_4,c}$).

Option 2: Procedure using default values

A default value is used by PP, i.e. $EF_{CH_4,y} = EF_{CH_4,default}$, which is verified to be in compliance with Tool "Project and leakage emissions from composting" (Version 02.0). As per the "Data / Parameter table 2" of this tool, $EF_{CH_4,default}$ is 0.002 tCH₄/t.

Determination of project emissions of nitrous oxide ($PE_{N_2O,y}$)

As per para 26 of Tool "Project and leakage emissions from composting" (Version 02.0), project emissions of nitrous oxide from composting ($PE_{N_2O,y}$) are determined as follows:

$$PE_{N_2O,y} = Q_y \times EF_{N_2O,y} \times GWP_{N_2O} \quad \text{Equation (8)}$$

Where

$PE_{N_2O,y}$	=	Project emissions of nitrous oxide from composting in year y (t CO ₂ e/yr)
Q_y	=	Quantity of waste composted in year y (t/yr)
$EF_{N_2O,y}$	=	Emission factor of nitrous oxide per tonne of waste composted valid for year y (t N ₂ O/t)
GWP_{N_2O}	=	Global Warming Potential of N ₂ O (t CO ₂ e / t N ₂ O)

Option 1: Procedure using monitored data

$EF_{N_2O,y}$ is determined based on measurements of the emissions during a composting cycle ($ECC_{N_2O,c}$).

Option 2: Procedure using default values

A default value is used by PP, i.e. $EF_{N_2O,y} = EF_{N_2O,default}$, which is verified to be in compliance with Tool "Project and leakage emissions from composting" (Version 02.0). As per the "Data / Parameter table 3" of this tool, the $EF_{N_2O,default}$ is 0.0002 t N₂O/t.

Determination of project emissions from run-off wastewater ($PE_{RO,y}$)

As per para 30 of Tool "Project and leakage emissions from composting" (Version 02.0), project emissions of methane from run-off wastewater ($PE_{RO,y}$) are calculated only for the case of co-composting. Moreover, if run-off wastewater is collected and re-circulated to the composting process. As justified above, the project activity uses a solid-liquid separation system to separate solids in swine manure from swine farms in Leiyang City for further producing organic solid fertilizers through aerobic composting. No co-composting wastewater was involved in the project activity. Therefore, $PE_{RO,y} = 0$.

✓ Project emissions from energy use for operating separation devices ($PE_{y,power}$)

As per para 35 of AMS-III.Y. (Version 4.0), project activity emissions from electricity or fossil fuel consumption ($PE_{y,power}$) are determined as per the procedures described in AMS-I.D. The energy consumption of all equipment/devices installed by the project activity, inter alia all equipment to separate solids (including energy used for spray drying and evaporation) shall be included. For project

activity emissions from fossil fuel consumption the emission factor for the fossil fuel shall be used ($\text{tCO}_2/\text{tonne}$). Local values are to be used, if local values are difficult to obtain, IPCC default values may be used. If recovered methane is used to power auxiliary equipment of the project it should be taken into account accordingly, using zero as its emission factor. That is to say, $\text{PE}_{y,\text{power}} = \text{PE}_{y,\text{electricity}} + \text{PE}_{y,\text{fuel}}$

For the project activity, since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively measured and recorded. Electricity consumption from the project activity has been used as $\text{EC}_{\text{PJ},\text{compost},y}$ in calculation of $\text{PE}_{\text{EC},y}$. Therefore, it was verified $\text{PE}_{y,\text{electricity}}$ has been included in $\text{PE}_{\text{EC},y}$.

Via site inspection, it was verified the solid separation system consumes only electricity during its operation, no fossil fuel was consumed. Therefore, it was verified $\text{PE}_{y,\text{fuel}} = 0$.

✓ **Project emissions from combustion of flocculants ($\text{PE}_{y,\text{floc,combustion}}$)**

As per para 36 of AMS-III.Y. (Version 4.0), in case of combustion of separated solids, the CO_2 emissions from the combustion of the non-biomass (i.e. fossil) carbon content of the flocculants shall be considered. In such a case, a default value of $0.2 \text{ tCO}_2\text{e}$ per tonne of flocculants may be used.

By checking FSR /14/, onsite inspection and interviewing with PP, it was verified no flocculants were used during solid separations. Therefore, $\text{PE}_{y,\text{floc,combustion}} = 0$.

✓ **Project emissions from incremental transportation of matters ($\text{PE}_{y,\text{trans}}$)**

As per para 29 of AMS-III.Y. (Version 4.0),

Incremental CO_2 emissions due to increased transportation ($\text{PE}_{y,\text{transp}}$):

- (i) Transportation of solids to sites where it will be treated further or gainfully used (within the project boundary);
- (ii) Transportation of solids from and to treatment facilities to storage sites (within the project boundary);
- (iii) Transportation of solids to disposal site.

Therefore, for the project activity $\text{PE}_{y,\text{transp}}$ includes the following components:

- a. Fresh manure slurry from the swine farms to the project site;
- b. Transportation of separated solids from the on-site solid separation system to the on-site composting system;
- c. Transportation of the composted solid fertilizers to local farms for land application.

For component (a), by checking Livestock Manure Treatment Agreement /69/, List of livestock farm /67/, interviewing PP during site visit, as well as onsite investigation, it was verified the swine farms involved in the project are from Leiyang City (county level). During site visit, checking nameplate of trucks, it was verified trucks used to transport manure slurry are with the minimum total mass of 8 tonnes.

For component (b), as justified above, for the project, all separated solids are composted on site without storage, destruction or land application, prior to any secondary use. During on-site inspection and interview, it was verified the solids from the separation system immediately enter the composting system. Usually the separated solids are transported by using wheelbarrows. Besides, via site inspection, it was verified the transportation distance between these two systems is minimal ($DT_y < 0.05\text{km/truck}$). Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.

For component (c), by checking Fertilizer Sales Record /82/, interviewing PP during site visit, as well as onsite investigation, it was verified the produced compost is entirely sold externally in Leiyang City (county level) and trucks used to transport solid fertilizers are with the minimum total mass of 10 tonnes.

As per para 37 of AMS-III.Y. (Version 4.0), project activity emissions from trucks for incremental transport activities of matters shall be calculated as follows:

$$PE_{y,transp} = \frac{Q_{y,transp}}{CT_y} \times DT_y \times EF_{CO_2} \quad \text{Equation (9)}$$

Where:

$Q_{y,transp}$	=	Quantity of matters transported in year y (t)
CT_y	=	Average truck capacity for transportation (t/truck)
DT_y	=	Average incremental distance for transportation of matters (km/truck)
EF_{CO_2}	=	CO ₂ emission factor for the fuel used for transportation (tCO ₂ /km, IPCC default values or local values)

The parameters used to calculate PE_y ex-ante is determined and justified as below:

Parameter	Value	Data sources
Q_y	52,500 tonnes	By checking FSR /14/, it is confirmed the project activity could treat approximately 52,500 tonnes of manure annually.
$EC_{PJ,compost,y}$	80 MWh	During site inspection and interviewing with PP, it was verified the monitored data for quantity of electricity consumed by composting ($EC_{PJ,compost,y}$) is not available since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively

		measured and recorded. Therefore, electricity consumption from the project activity is used as $EC_{PJ,compost,y}$ in calculation of $PE_{EC,y}$, which is conservative. By checking FSR /14/, it was verified electricity consumption from the project activity that involves all facilities and equipment within the project boundary is 80MWh.
$EF_{EF,composting,y}$	1.3 tCO ₂ e/MWh	Default value from Tool 05: "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (Version 03.0)
$TDL_{composting,y}$	20%	Default value from Tool 05 "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (Version 03.0)
$EF_{FC,default}$	0.0207 tCO ₂ /t	Default value from "Data / Parameter table 5" in Tool 13 "Project and leakage emissions from composting" (Version 02.0)
$EF_{CH_4,y}$	0.002 tCH ₄ /t	Default value from "Data / Parameter table 2" in Tool 13 "Project and leakage emissions from composting" (Version 02.0)
GWP_{CH_4}	28 tCO ₂ /tCH ₄	IPCC Fifth Assessment Report (AR5)
$EF_{N_2O,y}$	0.0002 tN ₂ O/t	Default value from "Data / Parameter table 3" in Tool 13 "Project and leakage emissions from composting" (Version 02.0)
GWP_{N_2O}	265 tCO ₂ /tN ₂ O	IPCC Fifth Assessment Report (AR5)
$Q_{y,transp}$	130,000 tonnes	By checking FSR /14/, it was verified fresh manure slurry transported from the swine farms to the project site and the solid fertilizer exported from the project site to local farmlands were 130,000 tonnes, including 100,000 tonnes fresh manure slurry transported from the swine farms to the project site and 30,000 tonnes solid fertilizer exported from the project site to local farmlands.
CT_y	8 tonnes/truck	During site visit, checking nameplate of trucks, it was verified trucks used to transport manure slurry are with the minimum total mass of 8 tonnes. By checking Fertilizer Sales Record /82/, it was verified trucks used to transport solid fertilizers are with the minimum total mass of 10 tonnes. PP applied 8 tonnes/truck for all the trucks, which is verified to be conservative.
DT_y	200 km	Considering swine farms and farmlands utilize the fertilizer productions are within Leiyang City (county level), DT_y is determined based on the farthest point from the project site within the boundary of Leiyang City (county level), which is verified to be conservative. By checking ArcMap App, it was verified the farthest point from the project site within the boundary of Leiyang City (county level) was located at 26.693° N and 113.177° E, which was 85.8 km far away from the project site as confirmed via Google Map, therefore the to and fro distance was verified to be 200km/truck.

EF_{CO_2}	0.00082 tCO ₂ /km	Please refer to section 3.4.8 of this report for detailed justification of value of this parameter.
$PE_{EC,y}$	125 tCO ₂ e	calculated via equation (5)
$PE_{FC,y}$	1,087 tCO ₂ e	calculated via equation (6)
$PE_{CH_4,y}$	2,940 tCO ₂ e	Calculated via equation (7)
$PE_{N_2O,y}$	2,783 tCO ₂ e	Calculated via equation (8)
$PE_{RO,y}$	0	No co-composting wastewater was involved in the project activity.
$PE_{y,ss}$ or $PE_{s,composting,y}$	6,935 tCO₂e	Calculated via equation (4)
$PE_{y,power}$	0	As justified above, For the project activity, since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively measured and recorded. Electricity consumption from the project activity has been used as $EC_{PJ,compost,y}$ in calculation of $PE_{EC,y}$. Therefore, it was verified $PE_{y,electricity} = PE_{EC,y}$. For the project activity, project emissions from fossil fuel consumption ($PE_{FC,y}$) has been included. Therefore, it was verified $PE_{y,fuel} = PE_{FC,y}$.
$PE_{y,floc,combustion}$	0	By checking FSR /14/, onsite inspection and interviewing with PP, it was verified no flocculants were used during solid separations.
$PE_{y,transp}$	2,665 tCO₂e	Calculated via equation (9)
PE_y	9,600 tCO₂e	Calculated via equation (2)

Leakage

As per paragraph 38 and 39 of AMS-III.Y (Version 4.0), if the solid separation technology is equipment transferred from another activity leakage effects are to be considered. When flocculants is used and if the flocculants includes ingredients that were manufactured and not a waste product, then leakage emissions must be addressed for that portion of the flocculants. A default emission factor of 7.9 tCO₂e/t may be applied based on the amount of the flocculants that is manufactured (i.e. excluding the portions that obtained from waste sources).

As justified above, the project separates solids from collected manure by applying solid separation technology installed at the project site. By checking Equipment Purchase Agreement /24//25/, it was verified these solid separation equipments were newly bought at the time of operation start, not transferred from another project activity. Besides, as verified, no flocculants have been used during composting. Therefore, leakage emissions of the project is not accounted for.

Emission Reductions

According to AMS-III.Y., Version 4.0, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (10)}$$

Where:

ER_y	=	Emission reductions in year y (t CO ₂ e/yr)
BE_y	=	Baseline emissions in year y (t CO ₂ e/yr)
PE_y	=	Project emissions in year y (t CO ₂ e/yr)
LE_y	=	Leakage emissions in year y (t CO ₂ e/yr)

Hence for the project activity, the estimated amount of GHG emission reductions (ER_y) ex ante is 57,307 tCO₂e during the crediting period from 28/03/2022 to 27/03/2029 , resulting in estimated average annual emission reductions of 57,307 tCO₂e.

Year	BE_y (tCO ₂ e)	PE_y (tCO ₂ e)	LE_y (tCO ₂ e)	ER_y (tCO ₂ e)
28/03/2022 to 31/12/2022	51,143	7,338	0	43,805
01/01/2023 to 31/12/2023	66,907	9,600	0	57,307
01/01/2024 to 31/12/2024	66,907	9,600	0	57,307
01/01/2025 to 31/12/2025	66,907	9,600	0	57,307
01/01/2026 to 31/12/2026	66,907	9,600	0	57,307
01/01/2027 to 31/12/2027	66,907	9,600	0	57,307
01/01/2028 to 31/12/2028	66,907	9,600	0	57,307
01/01/2029 to 27/03/2029	15,764	2,262	0	13,502
Total	468,349	67,200	0	401,149

The assessment team confirms that all data sources and assumptions are appropriate and calculations are correct, applicable to the project activity and will result in a conservative estimate of the emission reductions. In general, the assessment team is able to confirm the following:

- All assumptions and data used by the project participants are listed in the joint PD & MR (version 3.0 dated 06/09/2024) and/or supporting documents, including their references and sources;
- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the joint PD & MR (version 3.0 dated 06/09/2024);
- All values used in joint PD & MR (version 3.0 dated 06/09/2024) are considered reasonable in the context of the project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;

- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in joint PD & MR (version 3.0 dated 06/09/2024).

3.4.7 Methodology Deviations

No methodology deviations were applied for the project activity.

3.4.8 Monitoring Plan

The project applies the approved monitoring methodology - AMS-III.Y.: Methane avoidance through separation of solids from wastewater or manure treatment systems, Version 4.0. The selected monitoring methodology is applicable for the project. The monitoring plan is in accordance with the monitoring methodologies. The monitoring plan will give opportunity for real measurements of achieved emission reductions and contains principles and concepts on which it is based, operational and monitoring obligations of the project owner like resources involved in the monitoring process, training, support activities, calibration and data collection, quality assurance procedures, data management, electronic support tools.

Parameters determined ex-ante

Data / Parameter:	$B_{o,w,y}$
Data unit:	m ³ CH ₄ /kg-VS
Description:	Weighted methane-producing potential of the volatile solids separated by the project in year y
Source of data used:	Table 10.16 of Volume 4 Chapter 10 of 2019 Refinement to the 2006 IPCC Guidelines
Value applied:	0.45 m³CH₄/kg-VS (swine) According to the scientific publication database, i.e. www.cnki.net, it was verified no specific country or regional specific value in China are available. During site visit, as well as via checking Project Approval /15/, Environmental Impact Assessment (EIA) approval /17/, it was verified the project is located in China, Asia. By checking FSR /14/, list of Livestock Farms /67/ and through site inspection and interview, it was verified the swine farms are 100 percent market oriented with high level of capital input and high level of overall herd (flock) performance. Feed is purchased from local markets, swines are improved through breeding practices for commercial production and thereby the swine farms are with "High Productivity Systems". IPCC default values from Table 10.16 of Volume 4 Chapter 10 of 2019 Refinement to the 2006 IPCC Guidelines for "swine" in "Other Region" with

	"High Productivity Systems", i.e. $0.45 \text{ m}^3\text{CH}_4/\text{kg-VS}$ was applied for B_o .
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Data / Parameter:	UF_b
Data unit:	/
Description:	Model correction factor to account for model uncertainties
Source of data used:	AMS-III.Y (version 04.0)
Value applied:	0.94

Data / Parameter:	GWP_{CH_4}
Data unit:	/
Description:	Global warming potential of CH_4
Source of data used:	IPCC AR5
Value applied:	28

Data / Parameter:	GWP_{N_2O}
Data unit:	tCO_2e/tN_2O
Description:	Global warming potential of N_2O
Source of data used:	Default value from IPCC Fifth Assessment Report (AR5)
Value applied:	265

Data / Parameter:	D_{CH_4}
Data unit:	kg/m^3
Description:	CH_4 density
Source of data used:	AMS-III.Y (version 04.0)
Value applied:	0.67 (at 20°C and 1 atm pressure)

Data / Parameter:	$MS_{BI,i}$
Data unit:	/
Description:	Fraction of manure handled in the baseline anaerobic manure management

	system i (fraction, mass basis)
Source of data used:	AMS-III.Y (version 04.0)
Value applied:	1.0 By checking FSR /14/, Livestock Manure Treatment Agreement /69/ and List of livestock farm /67/, the livestock farms involved and their respective townships are confirmed. Also by checking FSR /14/ and via on-site inspection (photos of baseline treatment system have been taken by the assessment team) and interviewing livestock farmers, it was verified only one type of treatment system, i.e. uncovered anaerobic lagoon, was used in the baseline scenario of the livestock farms. Therefore, as per AMS-III.Y, the value of $MS_{BI,i}$ is “1.0”. By checking FSR/14/ and Project Approval /15/, it was verified FSR was completed in January 2020 and approved by Leiyang Development and Reform Bureau on 22/04/2020. Considering the project started operation on 28/03/2022, the assessment team verified the historical records are at least one year prior to the operation start date of the project activity and thereby is in compliance with para.22 of AMS III.Y, i.e. in the determination of baseline emissions, historical records of at least one year prior to the project implementation shall be used.

Data / Parameter:	$MCF_{b,i}$
Data unit:	/
Description:	Methane conversion factor for the baseline anaerobic manure management system
Source of data used:	Table 10.17(UPDATED) of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
Value applied:	73% By checking FSR/14/ and via on-site inspection (photos of baseline treatment system have been taken by the assessment team) and interviewing livestock farmers, it was verified only one type of treatment system, i.e. uncovered anaerobic lagoon, was used in the baseline scenario of the livestock farms, where animal manure was left to decay in uncovered anaerobic lagoon at the local swine farms. By checking FSR/14/ and Project Approval /15/, it was verified FSR was completed in January 2020 and approved by Leiyang Development and Reform Bureau on 22/04/2020. Considering the project started operation on 28/03/2022, the assessment team verified the historical records are at least one year prior to the operation start date of the project activity and thereby is in compliance with para.22 of AMS III.Y, i.e. in the determination of baseline emissions, historical records of at least one year prior to the project implementation shall be used. By checking public information /21/, it is confirmed the climate zone of project site belongs to Warm Temperate Moist. Therefore, default value of Warm Temperate Moist, i.e. 73%, sourced from Table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas

	Inventories Volume 4 Chapter 10 was applied.
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Data / Parameter:	EF_{EF,composting,y}
Data unit:	tCO ₂ e/ MWh
Description:	Combined margin emission factor for the grid in year y
Source of data used:	Tool 05 “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 03.0)
Value applied:	1.3 The project activity consumed electricity from the grid, which applies only only to project and/or leakage electricity consumption sources but not to baseline electricity consumption sources. As per section 6.2.1.1 of Tool 05, in this case, PP may use a conservative default value of 1.3 t CO ₂ /MWh.

Data / Parameter:	TDL_{compost,y}
Data unit:	/
Description:	Average technical transmission and distribution losses for providing electricity to composting in year y
Source of data used:	Tool 05 “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 03.0)
Value applied:	20% As per “Data / Parameter table 3” of Tool 05, PP can use as default values of 20% for project or leakage electricity consumption sources.

Data / Parameter:	EF_{FC,default}
Data unit:	tCO ₂ /t
Description:	Default emission factor for fossil fuel consumed by the composting activity per tonne of waste composted (wet basis)
Source of data used:	Tool 13 “Project and leakage emissions from composting” (Version 02.0)
Value applied:	0.0207

Data / Parameter:	EF_{CH4,y} = EF_{CH4,default}
Data unit:	tCH ₄ /t

Description:	Default emission factor of methane per tonne of waste composted (wet basis)
Source of data used:	Tool 13 “Project and leakage emissions from composting” (Version 02.0)
Value applied:	0.002 Default value from “Data / Parameter table 2” in Tool 13 “Project and leakage emissions from composting” (Version 02.0) was applied.

Data / Parameter:	$EF_{N2O,y} = EF_{N2O,default}$
Data unit:	tN ₂ O/t
Description:	Default emission factor of nitrous oxide per tonne of waste composted (wet basis)
Source of data used:	Tool 13 “Project and leakage emissions from composting” (Version 02.0)
Value applied:	0.0002 Default value from “Data / Parameter table 3” in Tool 13 “Project and leakage emissions from composting” (Version 02.0) was applied.

Data / Parameter:	DT_y
Data unit:	km/truck
Description:	Average incremental distance for transportation of matters
Source of data used:	Maximum road distance of incremental transportation obtained from the maps
Value applied:	200 Considering swine farms and farmlands utilize the fertilizer productions are within Leiyang City (county level), DT_y is determined based on the farthest point from the project site within the boundary of Leiyang City (county level), which is verified to be conservative. By checking ArcMap App, it was verified the farthest point from the project site within the boundary of Leiyang City (county level) was located at 26.693° N and 113.177° E, which was 85.8km far away from the project site as confirmed via Google Map. PP determined the the to and fro distance as 200km/truck (100km/truck *2), which was verified to be conservative.

Data / Parameter:	CT_y
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Data unit:	tonnes/truck
Description:	Average truck capacity for transportation
Source of data used:	Minimum capacity of transportation trucks
Value applied:	8 During site visit, checking nameplate of trucks, it was verified trucks used to transport manure slurry are with the minimum total mass of 8 tonnes. By checking Fertilizer Sales Record /82/, it was verified trucks used to transport solid fertilizers are with the minimum total mass of 10 tonnes. PP applied 8 tonnes/truck for all the trucks, which is verified to be conservative.

Data / Parameter:	EF _{CO2}														
Data unit:	tCO ₂ /km														
Description:	CO ₂ emission factor for the fuel used for transportation (IPCC default values or local values)														
Source of data used:	Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/ 2006 IPCC Guidelines for National Greenhouse Gas Inventories /85/														
Value applied:	0.00082 Value of this parameter was calculated via equation: EF_{CO2} = (V × ρ) × NCV × EF_{CO2,TJ} Where: <table><tr><td>V</td><td>=</td><td>Fuel consumption per 100 km (V) for diesel trucks, i.e. 0.0307 m³/100km, which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/</td></tr><tr><td>ρ</td><td>=</td><td>The default density for diesel oil, i.e. 0.84 t/m³, which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/</td></tr><tr><td>NCV</td><td>=</td><td>The default net calorific value of diesel oil, i.e. 43 TJ/Gg, which was sourced from Table 1.2 of the Introduction Chapter, Volume 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories /85/</td></tr><tr><td>EF_{CO2,TJ}</td><td>=</td><td>The road transportation default CO₂ emission factor of</td></tr></table>			V	=	Fuel consumption per 100 km (V) for diesel trucks, i.e. 0.0307 m ³ /100km, which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/	ρ	=	The default density for diesel oil, i.e. 0.84 t/m ³ , which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/	NCV	=	The default net calorific value of diesel oil, i.e. 43 TJ/Gg, which was sourced from Table 1.2 of the Introduction Chapter, Volume 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories /85/	EF _{CO2,TJ}	=	The road transportation default CO ₂ emission factor of
V	=	Fuel consumption per 100 km (V) for diesel trucks, i.e. 0.0307 m ³ /100km, which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/													
ρ	=	The default density for diesel oil, i.e. 0.84 t/m ³ , which was sourced from Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) /86/													
NCV	=	The default net calorific value of diesel oil, i.e. 43 TJ/Gg, which was sourced from Table 1.2 of the Introduction Chapter, Volume 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories /85/													
EF _{CO2,TJ}	=	The road transportation default CO ₂ emission factor of													

		diesel oil, i.e. 74,100 kgCO ₂ /TJ, which was sourced from Table 3.2.1 of Chapter 3, Volume 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories /85/
Based on the above justifications, EF _{CO2} is verified to be 0.00082 tCO ₂ /km.		

Parameters determined ex-post

Parameter	Description	Monitoring procedure
M _{ss,y}	Mass (dry matter basis) of total separated solids in year y	As per para 42 (a) of AMS-III.Y (version 04.0), mass of separated solids (M_{ss,y}), measured by direct weighing of all separated solids, and measuring its dry matter content through representative sampling. If the dry matter content of a sample is lower than the minimum value of 20%, no emission avoidance will be assigned to the amount of separated solids from which the sample is representative. During site visit, it was verified this parameter will be calculated via equation “Q_y (t) * 1000 kg *(1-moisture content (%))”. Q_y is measured continuously by direct weighing using the scale installed near the separation system and aggregated in monitoring records. The moisture content will be monitored through laboratory testing. Samples will be taken from separated solids of manure by staff from the project owner. Samples will be tested by oven and analytical balance. The monthly value of M_{ss,y} will be aggregated based on monitoring data of Q_y and the moisture content. The monitoring instruments will be calibrated periodically as per national standards, eg. Verification Regulation for Digital Indicating Weighing Instruments (JJG 539-2016) /50/ and Verification Regulation of Electronic Balance (JJG1036-2008) /60/. The ex-ante value of this parameter was sourced from FSR /14/. By checking FSR/14/ and Project Approval /15/, it was verified FSR was completed in January 2020 and approved by Leiyang Development and Reform Bureau on 22/04/2020. Considering the project started operation on 28/03/2022, the assessment team verified the historical records are at least one year prior to the operation start date of the project activity and thereby is in compliance with para.22 of AMS III.Y, i.e. in the determination of baseline emissions, historical records of at least one year prior to the project implementation shall be used.

VS _{ss,y}	Volatile solids content of the separated solids in year y on a dry matter basis	During site visit, it was verified this parameter was determined as per Annex 2: Method for determination of Volatile Solids in animal waste of AM0073 version 01. The determination method was: Place Total Solids residue in furnace at 600°C for at least 1 hour. Volatile Solids are determined from weight difference of total and Fixed Solids. $\text{Volatile matter(drybasis)} = \frac{W_2 - W_f}{W_2 - W_1}$ Where W₁ is the weight of sample container, W₂ is combined weight of the sample container and oven dried sample, W_f is the combined constant weight of the sample container and sample after heating at 600°C. This parameter will be monitored based on daily measurement in accordance with the above requirements and equation. The ex-ante value of this parameter was sourced from FSR /14/. By checking FSR/14/ and Project Approval /15/, it was verified FSR was completed in January 2020 and approved by Leiyang Development and Reform Bureau on 22/04/2020. Considering the project started operation on 28/03/2022, the assessment team verified the historical records are at least one year prior to the operation start date of the project activity and thereby is in compliance with para.22 of AMS III.Y, i.e. in the determination of baseline emissions, historical records of at least one year prior to the project implementation shall be used.
Q _y	Quantity of waste composted in year y (wet basis)	This parameter will be measured daily by direct weighing using the scale installed near the separation system and aggregated in monitoring records, which is in compliance with para 15 of Tool13 version 02.0. The scale will be calibrated regularly as per relevant national regulations, eg. Verification Regulation for Digital Indicating Weighing Instruments (JJG 539-2016) /50/, by qualified third party.
EC _{PJ,compost,y}	Quantity of electricity consumed by the project electricity consumption from composting in year y	As justified above, the monitored data for quantity of electricity consumed by composting (EC_{PJ,compost,y}) is not available since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively measured and recorded. Therefore, electricity consumption from the project activity (EC_{PJ,project,y}) is used as EC_{PJ,compost,y} in calculation of PE_{EC,y}, which is conservative. During site visit, it was verified this parameter will be continuously monitored by a main electricity meter installed at the project site. At the end of each month, the staff from the project owner will read the electricity meter and record the meter readings. The electricity

		purchase receipts will be used to crosscheck. The electricity meter will be calibrated periodically as per national standards, eg. "Verification Regulation of Electrical Meters for Measuring Alternating-current Electrical Energy (JJG596-2012)" /65/.
$N_{LT,y}$	Number of swine for the year y	As per AMS-III.D, this parameter was determined via equation below: $N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365} \right)$ During site visit, it was verified the number of days that swines alive on the farm ($N_{da,y}$) and the number of swines produced by the project activity ($N_{p,y}$) will be monitored and aggregated yearly by responsible staff on each swine farm and the monitoring results will be collected by PP. The consistency between the value and indirect information (records of sales, records of food purchases) will be assessed during each verification. Monitoring of this parameter is verified to be in compliance with para. 43(b) of AMS-III.Y.
$VS_{LT,y}$	Annual amount of volatile solids excreted by swine managed by the management system in year y	According to the scientific publication database, i.e. www.cnki.net, it was verified no specific country or regional specific value in China are available. $VS_{LT,y}$ is calculated via equation 10.22A in 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. During site visit and via checking Project Approval /15/ and EIA Approval /17/, it was verified the project is located in Leiyang City (county level), Hengyang City, Hunan Province, Asia. By checking FSR /14/, it was verified the swine farm is with High Productivity Systems. Therefore, PP applied average high PS value for swine in Asia from Table 10.13A (NEW) of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10, i.e. 4.3kg-VS/1000kg-animal mass/day, for $VS_{rate(T,P)}$, which is verified to be reasonable. PP adopts high PS value of TAM for swine in Region "Asia" from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4, Chapter 10, Table 10A.5(NEW), i.e. 69kg for $TAM_{(T,P)}$, which is verified to be reasonable. Therefore, $VS_{LT,y}$ is calculated to be: $4.3\text{kg}/1000\text{kg animal mass}/\text{day} \times 69\text{kg animal mass}/\text{hd} \times 365\text{days} = 108.30\text{kg}/\text{hd}/\text{y}.$ The consistency between the value and indirect information (records of sales, records of food purchases) will be assessed. For ex-post determination, default values will be updated annually from latest IPCC

		values to determine $VS_{LT,y}$.						
$Q_{y,transp}$	Quantity of matter transported in year y	Value of this parameter was calculated via equation below: $Q_{y,transp} = Q_{y,internal} + Q_{y,external}$ <table border="1"> <tr> <td>$Q_{y,internal}$</td><td>=</td><td>The transportation of separated solids from the on-site solid separation system to the on-site composting system. As justified above, all separated solids are composted on site without storage, destruction or land application, prior to any secondary use. During on-site inspection and interview, it was verified the solids from the separation system immediately enter the composting system. Usually the separated solids are transported by using wheelbarrows. Besides, via site inspection, it was verified the transportation distance between these two systems is minimal (< 0.05km/truck). Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.</td></tr> <tr> <td>$Q_{y,external}$</td><td>=</td><td>The transportation of the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands.</td></tr> </table> Therefore, it was verified $Q_{y,transp} = Q_{y,external}$. The transportation of the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands will both be measured daily by direct weighing using the scale installed near the separation system and aggregated in monitoring records, which is in compliance with AMS III.Y. The scale will be calibrated regularly as per relevant national regulations, eg. Verification Regulation for Digital Indicating Weighing Instruments (JJG 539-2016) /50/, by qualified third party.	$Q_{y,internal}$	=	The transportation of separated solids from the on-site solid separation system to the on-site composting system. As justified above, all separated solids are composted on site without storage, destruction or land application, prior to any secondary use. During on-site inspection and interview, it was verified the solids from the separation system immediately enter the composting system. Usually the separated solids are transported by using wheelbarrows. Besides, via site inspection, it was verified the transportation distance between these two systems is minimal (< 0.05km/truck). Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.	$Q_{y,external}$	=	The transportation of the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands.
$Q_{y,internal}$	=	The transportation of separated solids from the on-site solid separation system to the on-site composting system. As justified above, all separated solids are composted on site without storage, destruction or land application, prior to any secondary use. During on-site inspection and interview, it was verified the solids from the separation system immediately enter the composting system. Usually the separated solids are transported by using wheelbarrows. Besides, via site inspection, it was verified the transportation distance between these two systems is minimal (< 0.05km/truck). Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.						
$Q_{y,external}$	=	The transportation of the import of fresh manure slurry from the swine farms to the project site, as well as the export of the solid fertilizer from the project site to the local farmlands.						

Furthermore, the operational conditions of the baseline manure management as required by para 41 of AMS III.Y version 4:

Condition	Description of the project
Animals confinement management status	By checking FSR /14/, it was verified swine have been managed in confined conditions, entailing the use of enclosed building structures equipped with ventilation systems, lighting facilities, and temperature control devices to ensure good air quality, appropriate temperatures, and adequate lighting conditions. Cement floors or manure leakage boards are commonly featured to provide a neat environment. Swine farms are equipped with fire prevention equipment, emergency exits, and surveillance cameras. The same has been confirmed via interviews with the PP and local farmers and on-site investigation.
Anaerobic lagoon overflow discharge status	By checking FSR /14/, it was verified swine manure from anaerobic lagoons were ultimately utilized for land application in Leiyang City (County Level), rather than being discharged into river/lake/sea. The same has been confirmed via interviews with the PP and local farmers, checking historical photographs of the anaerobic lagoons /76/ and Operation Records of baseline lagoons /79/. Similar statuses were also observed for surrounding swine farms (not within project scope) through on-site investigation.
Solids retention time and removal operation interval	By checking FSR /14/ and public information /75/, it was verified since the decomposed manure eventually applied to surrounding farmlands, farmers adhered to a schedule of removing solids from the anaerobic lagoons every 6 months in minimum to ensure thorough decomposition, safeguarding against adverse effects on the crops. The same has been confirmed via interviews with the PP and local farmers, and checking Operation Records of Baseline Lagoons /79/.
Solids separation facilities quality control	Via site visit, it was verified a quality control plan /80/ has been developed by the PP. It includes regular inspections and maintenance, dedicated on-site monitoring by trained personnel, and prompt repairs as needed. Regular inspections will ensure all equipment components are functioning optimally, while dedicated on-site monitoring by trained personnel will provide real-time oversight of operational parameters. Any deviations or issues detected will be addressed promptly through timely repairs to ensure uninterrupted and efficient operation of the solids separation facilities. The same has been confirmed via interviews with the PP and workers, checking Equipment Maintenance Records /78/ dating back to the start of the project.

Based on the on-site visit and interviewed with staff from Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd., The assessment team confirmed that monitoring parameter has been correctly described in the joint PD & MR (version 3.0 dated 06/09/2024) and in compliance with the methodology AMS-III.Y. Version 4.0, CDM Validation and Verification Standard for project activities version 03.0, and VCS standard Version 4.5.

3.5 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

4.1.1 Project Implementation Status

The selected monitoring period 28/03/2022 to 30/09/2023 is the 1st monitoring period of the project, which is within the crediting period 28/03/2022 to 27/03/2029 .

The control system at the project site is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records /34/ and interview with staff from project owner, CTI can confirm that no serious malfunction happened and the project site was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including monitoring, recording and reporting was verified to be in place and their implementation was confirmed by interview with the key operators and observing the operation.

Through visual inspection and document review through site visit, CTI was able to confirm that:

- the project implementation is in accordance with the project description contained in the Joint PD & MR (version 3.0 dated 06/09/2024) and monitoring plan.
- all physical features of the proposed project activity including data collection systems and storage systems have been implemented in accordance with the Joint PD & MR (version 3.0 dated 06/09/2024) and monitoring plan.
- the monitoring plan in the Joint PD & MR (version 3.0 dated 06/09/2024) is in accordance with the approved methodology applied by the project activity, i.e. AMS-III.Y. (Version 4.0).
- The monitoring has been carried out in accordance with the monitoring plan contained in the Joint PD & MR (version 3.0 dated 06/09/2024).
- all parameters stated in the monitoring plan are monitored and reported appropriately.

All parameters required to be monitored by the monitoring plan as per the monitoring methodology AMS-III.Y. (Version 4.0) and the management system were assessed during the site visit. The joint PD & MR lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, aggregation, recording, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

4.1.2 Parameters monitored ex post

In this monitoring period, the parameters monitored ex post were justified as follows:

- **$M_{ss,y}$: Mass (dry matter basis) of total separated solids in year y**

As per methodology and the monitoring plan, this parameter was calculated via equation " $Q_y(t) * 1000 \text{ kg} * (1 - \text{moisture content}(\%))$ ".

The detailed assessment of parameter Q_y has been included in the following text.

For moisture content, by checking Raw data of Solid Liquid Separation Record /45/ and via site investigation, it was verified after separation, three solid manure samples were taken daily for testing in laboratory by laboratory oven and analytical balance. For each of the three manure samples, test has been done via analytical balance, laboratory oven and muffle furnace everyday to ensure the accuracy of the testing result. The daily testing result, i.e. average value of these 3 testing results, has been recorded in Raw data of Solid Liquid Separation Record /45/ and monthly mean was calculated. Therefore, the assessment team verified the monitoring of moisture content is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Solid Liquid Separation Record /45/. All the data have been verified by CTI to be consistent. It is confirmed the value of $M_{ss,y}$ of this monitoring period was:

Period	Value (kg)
28/03/2022 to 31/12/2022	16,093,939
01/01/2023 to 30/09/2023	15,184,739
Total	31,278,678

● **$VS_{ss,y}$: Volatile solids content of the separated solids in year y on a dry matter basis**

By checking Solid Liquid Separation Record /45/ and via site investigation, it was verified after separation, three solid manure samples were taken daily for testing in laboratory by laboratory oven, analytical balance and muffle furnace. For each of the three manure samples, test has been done via analytical balance, laboratory oven and muffle furnace everyday to ensure the accuracy of the testing result. The daily testing result, i.e. average value of these 3 testing results, has been recorded in Raw data of Solid Liquid Separation Record /45/ and monthly mean was calculated. Therefore, the assessment team verified the monitoring of this parameter is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Solid Liquid Separation Record /45/. By checking the published research article “Research progress on bio-fuel prepared from livestock manure by hydrothermal treatment” /58/, it was verified the volatile solids content of swine manure is 55.30% - 69.84%. Therefore, the measured data of volatile solids content of separated solids of the project activity is deemed as reasonable by the assessment team. All the data have been verified by CTI to be consistent. It is confirmed the value of $VS_{ss,y}$ of this monitoring period was:

Period	Value (kg/kg)
28/03/2022 to 31/12/2022	0.55
01/01/2023 to 30/09/2023	0.56

● **Q_y : Quantity of waste composted in year (wet basis)**

By checking Solid Liquid Separation Record /45/ and via site investigation, it was verified after the collected manure was separated, this parameter was measured by an onsite scale everyday and the daily value was aggregated monthly, which is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Solid Liquid Separation Record /45/. All the data have been verified by CTI to be consistent. It is confirmed the value of Q_y of this monitoring period was:

Period	Value (t)
28/03/2022 to 31/12/2022	40,456.29
01/01/2023 to 30/09/2023	38,099.20
Sum	78,555.49

- **$EC_{PJ,compost,y}$: Quantity of electricity consumed by the project electricity consumption from composting in year y**

During site visit, it was verified the monitored data for quantity of electricity consumed by composting ($EC_{PJ,compost,y}$) is not available since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively measured and recorded. Therefore, electricity consumption from the project activity ($EC_{PJ,project,y}$) is used as $EC_{PJ,compost,y}$ in calculation of project emissions, which is conservative.

By checking Electricity Meter Reading Record /46/ and via site investigation, it was verified this parameter was continuously measured by a electricity meter installed at the project site, which is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Electricity Meter Reading Record /46/, which has been crosschecked with that in Electricity Purchase Receipts /47/. All the data have been verified by CTI to be consistent. It is confirmed the value of $EC_{PJ,compost,y}$ of this monitoring period was:

Period	Value (MWh)
28/03/2022 to 31/12/2022	12.219
01/01/2023 to 30/09/2023	9.573
Sum	21.792

- **$N_{LT,y}$: Number of swine for the year y**

As per AMS-III.D, this parameter was determined via equation below:

$$N_{LT,y} = N_{da,y} \times \left(\frac{N_{p,y}}{365} \right)$$

For $N_{da,y}$ and $N_{p,y}$: Via interview PP, it is confirmed an appointed staff from each swine farm is responsible for recording and aggregating annual operation records, PP collected the monitoring results from the farms and aggregate Yearly Inventory Statistics /84/, which is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Yearly Inventory Statistics /84/, which has been crosschecked with Statement about Swine Feed Purchase Statistics /87/. It is verified that the number of days animal is alive in the farm during this monitoring period and number of animals produced during this monitoring period is reasonable. All the data have been verified by CTI to be consistent.

It is confirmed that value of $N_{da,y}$ of this monitoring period was:

Period	Value (day)
28/03/2022 to 31/12/2022	279
01/01/2023 to 30/09/2023	273

It is confirmed that value of $N_{p,y}$ of this monitoring period was:

Period	Value (head)
28/03/2022 to 31/12/2022	84,558
01/01/2023 to 30/09/2023	80,286

Based on the above justifications, it was verified value of $N_{LT,y}$ of this monitoring period was:

Period	Value (head)
28/03/2022 to 31/12/2022	64,634
01/01/2023 to 30/09/2023	60,049

- **$VS_{LT,y}$: Annual amount of volatile solids excreted by swine managed by the management system in year y**

The default values used to calculate $VS_{LT,y}$ was sourced from latest version IPCC, i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10, for this monitoring period.

$VS_{LT,y}$ is calculated via equation 10.22A in 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, i.e. $VS_{(T,P)} = VS_{rate(T,P)} \times \left(\frac{TAM_{T,P}}{1000} \right) \times 365$.

PP applied high PS value for swine in Asia from Table 10.13A (NEW) of 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10, i.e. 4.3kg-VS/1000kg-animal mass/day, for $VS_{rate(T,P)}$, which is verified to be reasonable. PP adopts high PS value of TAM for swine in Region "Asia" from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4, Chapter 10, Table 10A.5, i.e. 69kg for $TAM_{(T,P)}$, which is verified to be reasonable. Therefore, $VS_{LT,y}$ is calculated to be: 4.3kg-VS/1000kg-animal mass/day * 69kg/1000*365 = 108.30 kg-VS/yr.

- **$Q_{y,transp}$: Quantity of matter transported in year y**

Value of this parameter was calculated via equation below:

$$Q_{y,transp} = Q_{y,internal} + Q_{y,external}$$

For $Q_{y,internal}$: the transportation of separated solids from the on-site solid separation system to the on-site composting system. As justified above, all separated solids are composted on site without storage, destruction or land application, prior to any secondary use. During on-site inspection and interview, it was verified the solids from the separation system immediately enter the composting system. Usually the separated solids are transported by using wheelbarrows. Besides, via site inspection, it was verified the transportation distance between these two systems is minimal (< 0.05km/truck). Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.

For $Q_{y,external}$: by checking Stock in and Out Transportation Records /83/ and via site investigation, it was verified fresh manure slurry transported from the swine farms to the project site and the solid

fertilizer exported from the project site to local farmlands were monitored. Value of this parameter was measured by an onsite scale everyday and the daily value was aggregated monthly, which is in compliance with methodology and the monitoring plan.

Data used for ER calculation was sourced from Stock in and Out Transportation Records /83/. All the data have been verified by CTI to be consistent. It is confirmed value of $Q_{y,transp}$ of this monitoring period was:

Period	$Q_{y,internal}$ (tonnes)	$Q_{y,external}$ (tonnes)
28/03/2022 to 31/12/2022	0	102,316
01/01/2023 to 30/09/2023	0	86,011
Sum	0	188,327

Monitoring equipment and calibration

The monitoring equipments have been calibrated periodically as per the relevant industrial standard by the qualified third party to ensure the monitoring equipments' accuracy and in good conditions. The relevant information of monitoring equipments' calibration /48//56//57/ is listed as below.

Scale ($M_{ss,y}$, Q_y and $Q_{y,transp}$)	
Type	FSG-30-QC
Serial No.	20210818111
Accuracy	III
Calibration frequency	Annually
Date of calibration	21/03/2022 21/03/2023
Date of Validity	20/03/2023 20/03/2024
Analytical balance ($M_{ss,y}$ and $VS_{ss,y}$)	
Type	FA2104
Serial No.	SHP0200578128
Accuracy	1mg
Calibration frequency	Annually
Date of calibration	21/03/2022 21/03/2023
Date of Validity	20/03/2023 20/03/2024
laboratory oven ($M_{ss,y}$ and $VS_{ss,y}$)	
Type	DHG-9050B
Serial No.	15041529
Accuracy	$\pm 1^\circ\text{C}$

Calibration frequency	-
Date of calibration	-
Date of Validity	-
Muffle furnace ($M_{ss,y}$ and $VS_{ss,y}$)	
Type	SX-4-10
Serial No.	-
Accuracy	-
Calibration frequency	-
Date of calibration	-
Date of Validity	-
Electricity meter ($EC_{PJ,compost,y}$)	
Type	DSZ331
Serial No.	00000025117515
Accuracy	0.5
Calibration frequency	Annually
Date of calibration	24/01/2022 24/01/2023
Date of Validity	23/01/2023 23/01/2024

The scale was calibrated by qualified third party, i.e. Hengyang County Quality Inspection and Metrological Verification Center. Calibration records and accreditation certificates /48//49/ have been verified by the assessment team. By checking the calibration reports, CTI found the scale has been calibrated every year, which satisfied with national or industry regulations and standards (i.e. Verification Regulation for Digital Indicating Weighing Instruments (JJG 539-2016) /50/) and is in line with methodology and the monitoring plan. Furthermore, the calibrations of these monitoring instruments are verified to be valid for the whole reporting period.

The analytical balance was calibrated by a qualified third party - Hengyang County Quality Inspection and Metrological Verification Center. Calibration records and accreditation certificates /56//49/ have been verified by the assessment team. By checking the calibration reports, CTI found the analytical balance has been calibrated annually as per "Verification Regulation of Electronic Balance (JJG1036-2008)" /60/, which is in line with the methodology and the monitoring plan. Furthermore, the calibrations of these monitoring instruments are verified to be valid for the whole reporting period.

The electricity meter was calibrated by a qualified third party - Hengyang County Quality Inspection and Metrological Verification Center. Calibration records and accreditation certificates /57//49/ have been verified by the assessment team. By checking the calibration reports, CTI found the electricity meter has been calibrated annually as per "Verification Regulation of Electrical Meters for Measuring Alternating-current Electrical Energy (JJG596-2012)" /65/, which is in line with the methodology and

the monitoring plan. Furthermore, the calibrations of these monitoring instruments are verified to be valid for the whole reporting period.

By checking “Announcement of the State Administration for Market Regulation on Issuing the Catalogue of Measuring Instruments for Compulsory Management” /64/ issued by the State Administration for Market Regulation on 23/10/2019, it was verified the measuring instruments listed in the “catalogue” should be subject to compulsory verification in use, and other measuring instruments will no longer be subject to compulsory verification. The laboratory oven and muffle furnace were not listed in the “catalogue” of this announcement for mandatory calibration and thereby not compulsorily calibrated. Therefore, no calibration of laboratory oven and muffle furnace were thereby accepted by the assessment team.

Data management and control

The project owner is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the VER project management and monitoring manual /29/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff and check records /29//34/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

4.1.3 Calculation of emission reductions

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

Baseline emissions

Based on parameters justified in section 3.4.8, section 4.1.2 and equation (1) in this report, baseline emissions have been calculated in ER calculation sheet and verified by the assessment team. it is confirmed BE_y was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
B _{0,w,y} (m ³ CH ₄ /kg-VS)	0.45	0.45
M _{ss,y} (kg)	16,093,939	15,184,739
V _{ss,y} (kg/kg)	0.55	0.56

GWP _{CH4} (tCO ₂ e/tCH ₄)	28	28
D _{CH4} (kg/m ³)	0.67	0.67
UF _b	0.94	0.94
MCF _j (%)	73%	73%
MCF _{b,i}	100%	100%
BE_y (tCO₂e)	51,276	49,259

Project emissions

As justified above, $PE_y = PE_{y,ss} + PE_{y,power} + PE_{y,floc,combustion} + PE_{y,transp}$

For $PE_{y,ss}$, it is calculated via equation: $PE_{y,ss} = PE_{s,composting,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH4,y} + PE_{N2O,y} + PE_{RO,y}$

$PE_{EC,y}$ is calculated based on parameters justified in section 3.4.8, section 4.1.2, equation (5) in this report. $PE_{EC,y}$ have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed $PE_{EC,y}$ was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
EC _{PJ,compost,y} (EC _{PJ,project,y})	12.219	9.573
EF _{EF,composting,y} (tCO ₂ /MWh)	1.3	1.3
TDL _{composting,y}	20%	20%
PE_{EC,y} (tCO₂e)	20	15

$PE_{FC,y}$ is calculated based on parameters justified in section 3.4.8, section 4.1.2 and equation (6) in this report. $PE_{FC,y}$ have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed $PE_{FC,y}$ was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
Q _y (tonnes/year)	40,456.29	38,099.20
EF _{FC,default} (tCO ₂ /t)	0.0207	0.0207
PE_{FC,y} (tCO₂e)	838	789

$PE_{CH4,y}$ is calculated based on parameters justified in section 3.4.8, section 4.1.2 and equation (7) in this report. $PE_{CH4,y}$ have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed $PE_{CH4,y}$ was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
Q _y (tonnes/year)	40,456.29	38,099.20
EF _{CH4,y} (tCO ₂ /t)	0.002	0.002
GWP _{CH4} (tCO ₂ e/tCH ₄)	28	28
PE_{CH4,y} (tCO₂e)	2,266	2,134

$PE_{N2O,y}$ is calculated based on parameters justified in section 3.4.8, section 4.1.2 and equation (8) in this report. $PE_{N2O,y}$ have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed $PE_{N2O,y}$ was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
Q_y (tonnes/year)	40,456.29	38,099.20
$EF_{N2O,y}$ (tCO ₂ /t)	0.0002	0.0002
GWP_{CH4} (tCO ₂ e/tCH ₄)	265	265
$PE_{N2O,y}$ (tCO₂e)	2,145	2,020

Therefore, $PE_{y,ss}$ is calculated to be:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
$PE_{EC,y}$ (tCO ₂ e)	20	15
$PE_{FC,y}$ (tCO ₂ e)	838	789
$PE_{CH4,y}$ (tCO ₂ e)	2,266	2,134
$PE_{N2O,y}$ (tCO ₂ e)	2,145	2,020
$PE_{RO,y}$ (tCO ₂ e)	0	0
$PE_{y,ss}$ (tCO₂e)	5,269	4,958

For $PE_{y,power}$, as justified above, for the project activity, since electricity consumption from the project activity that involves all facilities and equipment within the project boundary is collectively measured and recorded. Electricity consumption from the project activity has been used as $EC_{PJ,compost,y}$ in calculation of $PE_{EC,y}$. Therefore, it was verified $PE_{y,electricity}$ has been included in $PE_{EC,y}$.

Via site inspection, it was verified the solid separation system consumes only electricity during its operation, no fossil fuel was consumed. Therefore, it was verified $PE_{y,fuel} = 0$.

For $PE_{y,floc,combustion}$, by checking FSR /14/, onsite inspection and interviewing with PP, it was verified no flocculants were used during solid separations. Therefore, $PE_{y,floc,combustion} = 0$.

$PE_{y,transp}$ is calculated based on parameters justified in section 3.4.8, section 4.1.2 and equation (7) in this report. $PE_{CH4,y}$ have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed $PE_{CH4,y}$ was calculated as follows:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
$Q_{y,transp}$ (tonnes)	102,316	86,011
CT_y (tonnes/truck)	8	8
DT_y (km/truck)	200	200
EF_{CO2} (tCO ₂ /km)	0.00082	0.00082
$PE_{y,transp}$ (tCO₂e)	2,098	1,764

Therefore, PE_y is calculated to be:

Period	28/03/2022~31/12/2022	01/01/2023~30/09/2023
PE _{y,ss} (tCO ₂ e)	5,269	4,958
PE _{y,power} (tCO ₂ e)	0	0
PE _{y,floc,combustion} (tCO ₂ e)	0	0
PE _{y,transp} (tCO ₂ e)	2,098	1,764
PE _y (tCO ₂ e)	7,367	6,722

Leakage

As justified in section 3.4.6, leakage emissions of the project is not accounted for, i.e. LE_y = 0.

Emission reductions

ER_y is calculated based on equation (11), and have been calculated in ER calculation sheet and verified by the assessment team. It is confirmed ER_y = BE_y - PE_y - LE_y = 88,210 tCO₂e

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates ex-ante

The emission reductions claimed are 88,210 tCO₂e in this monitoring period. Compared with expected emission reductions 86,667 tCO₂e (57,307 tCO₂e / 365 days * 552 days) calculated ex-ante, the reported emission reductions in this monitoring period are 1.78% less than that expected, which is considered to be in the reasonable variation range.

Verified GHG emission reductions and removals from 28/03/2022 to 30/09/2023:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
28/03/2022~31/12/2022	51,276	7,367	0	43,909
01/01/2023~30/09/2023	49,259	4,958	0	44,301
Total	100,535	12,325	0	88,210

Year	Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
28/03/2022~30/09/2023	86,667	88,210	-1.78%	The difference is slight and thereby the variation is deemed as reasonable by the assessment team.
Total	86,667	88,210	-1.78%	

CTI is able to confirm that the actual emission reductions reported in this monitoring period are reasonable and appropriate. CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. And CTI is able to confirm that GHG emission reductions and removals have been quantified correctly in accordance with the monitoring plan and applied methodology for this verification period.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the joint PD & MR /1/ and ER Calculation Spreadsheet /3/ of this monitoring period and are consistent with the applied methodology - AMS-III.Y.: Methane avoidance through separation of solids from wastewater or manure treatment systems, Version 4.0 and the monitoring plan contained in the joint PD & MR. The parameters are complete in this monitoring period.

After verifying the reported figures with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation Spreadsheet of this monitoring period and joint PD & MR. the verification process for the same has been clearly described above in section 4 of the report.

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CTI concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

5 VALIDATION AND VERIFICATION CONCLUSION

CTI Certification Co., Ltd (CTI) has performed the joint validation and verification of the project activity "Leiyang Hexi Animal Manure Composting Project" in China (VCS Project ID: 4794) for the monitoring period 28/03/2022 to 30/09/2023 within the 1st crediting period 28/03/2022 to 27/03/2029 .

The review of the project design documentation and the subsequent follow-up interviews have provided CTI with sufficient evidence to determine the fulfilment of stated criteria.

In the course of the validation and verification, 1 Clarification Requests (CLs), 2 Corrective Action Requests (CARs) and no Forwarded Action Requests (FARs) was raised.

The review of the project design documentation and additional documents related to baseline and monitoring methodology and subsequent background investigation have provided the CTI with sufficient evidence to validate the fulfilment of the stated criteria. In detail the conclusions can be summarized as follows:

- the baseline scenario is correctly defined as per the applied methodology and relate tools;
- the project is additional;
- all data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;
- a reasonable level of assurance has been applied.
- the monitoring plan in the validated PD is as per the applied methodology AMS-III.Y. (Version 4.0);
- the project has been implemented and operated as per the joint PD & MR;
- the project complies with the validation criteria for projects set out in VCS Version 4.5;
- The calculation of the ex ante emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 401,149 tCO₂e are most likely to be achieved within the first crediting period started from 28/03/2022.

It is CTI's responsibility to provide an independent validation statement on the Project's compliance with the VCS Program requirements as set out in the VCS Program Rules. The conclusions of this report show that the project, as it was described in the project description contained in the joint PD & MR, is in line with all criteria applicable for the validation against the VCS standard Version 4.5 without any qualifications or limitations.

Year	Estimated net GHG emission reductions or removals (tCO ₂ e)
28/03/2022 to 31/12/2022	43,805
01/01/2023 to 31/12/2023	57,307

01/01/2024 to 31/12/2024	57,307
01/01/2025 to 31/12/2025	57,307
01/01/2026 to 31/12/2026	57,307
01/01/2027 to 31/12/2027	57,307
01/01/2028 to 31/12/2028	57,307
01/01/2029 to 27/03/2029	13,502
Total estimated ERs	401,149
Total number of crediting years	7
Average annual ERs	57,307

The verification is based on the baseline and monitoring methodology AMS-III.Y. (Version 4.0), and the joint PD & MR (version 3.0 dated 06/09/2024). The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the joint PD & MR. Our verification approach was based on the requirements as defined under the applicable VCS Version 4.5 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the monitoring plan in joint PD & MR is as per the applied methodology;
- the monitoring complies with the monitoring plan in the joint PD & MR;
- the joint PD & MR and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Version 4.5 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS

Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Leiyang Hexi Animal Manure Composting Project" for the monitoring period 28/03/2022 to 30/09/2023 are fairly stated in the joint PD & MR (version 3.0 dated 06/09/2024). The GHG emission reductions were calculated correctly on the basis of the approved methodology AMS-III.Y. (Version 4.0) and the monitoring plan contained in the joint PD & MR (version 3.0 dated 06/09/2024).

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Verification period: From 28/03/2022 to 30/09/2023

Verified GHG emission reductions and removals in the above verification period:

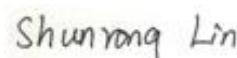
Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
28/03/2022~31/12/2022	51,276	7,367	0	43,909
01/01/2023~30/09/2023	49,259	4,958	0	44,301
Total	100,535	12,325	0	88,210



Mr. Feng Tian

Team Leader

10/09/2024



Ms. Lin Shunrong

Technical Reviewer

10/09/2024

APPENDIX 1: ABBREVIATIONS

Abbreviations	Full texts
AFOLU	Agriculture, Forestry, and Other Land Use
BM	Build Margin
CDM	Clean Development Mechanism
CME	Coordinating/managing entity
CER	Certified Emission Reduction
CM	Combined Margin
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DORs	Daily Production Records
EB	Executive Board
GHG	Greenhouse gas(es)
GS	Gold Standard
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETS	Emission Trading Scheme
FAR	Forward Action Request
GHG	Greenhouse gas(es)
IPCC	Intergovernmental Panel on Climate Change
MRRs	Monthly Meter Reading Records
MP	Monitoring Plan
PA	Project Activity
PD	Project Description
PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance / Quality Control
UNFCCC	United Nations Framework Convention on Climate Change
VVS	Validation and Verification Standard
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit
VVB	Validation/Verification Body

APPENDIX 2: <COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS>

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

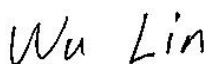
Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	√	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 8: Mining/mineral production	TA 8.1: Mining/mineral production
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN



Technical Competent Manager

Shenzhen, 25/10/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 3: Energy demand	TA 3.1: Energy demand
SS 13: Waste handling and disposal	TA 13.1: Solid waste and wastewater
	TA 13.2: Manure
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 25/10/2022

Mr. Qinghua DAI

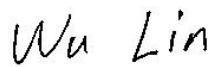
Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	-	-	-	-	-	√

Scope	Technical Area
SS 13: Waste handling and disposal	TA 13.2: Manure
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:



Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2024

APPENDIX 3: <REFERENCE LIST>

1. Joint project design and monitoring report, version 1.0 dated 30/10/2023, version 2.0 dated 07/06/2024, version 3.0 dated 06/09/2024
2. ER calculation spreadsheet for ex-ante estimation and for monitoring period from 28/03/2022 to 30/09/2023
3. VCS Standard Version 4.5, issued on 19/09/2019, updated on 11/12/2023
4. VCS Program Guide Version 4.4, issued on 19/09/2019, updated on 29/08/2023
5. VCS Joint Project Description & Monitoring Report Template version 4.2
6. CDM Validation and Verification Standard for project activities version 03.0
7. CDM Project Standard for project activities version 03.0
8. CDM project cycle procedure for project activities version 03.0
9. AMS-III.Y.: Methane avoidance through separation of solids from wastewater or manure treatment systems, Version 4.0
10. AMS-III.F: Avoidance of methane emissions through composting, version 12.0
AMS-I.D: Grid connected renewable electricity generation, version 18.0
11. Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (Version 03.0)
12. Tool 13: Project and leakage emissions from composting (Version 02.0)
13. Tool 21: Demonstration of additionality of small-scale project activities (Version 13.1)
14. Feasibility Study Report (FSR) compiled by Leiyang city Cailun modern agricultural science and technology Park development Co., LTD in April 2019 and revised in January 2020 due to productivity change of organic fertilizers
15. Project Approval of the project activity issued by Leiyang Development and Reform Bureau initially on 16/07/2019 and revised on 22/04/2020
16. Environmental Impact Assessment Report compiled by Chongqing Fengda Environmental Impact Assessment Co., LTD in July 2020
17. Environmental Impact Assessment (EIA) Approval issued by Leiyang City Ecological Environmental Bureau on 15/01/2021
18. Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022 issued by Ministry of Ecology and Environment of P.R.China
19. Notice on Strengthening Enterprise Greenhouse Gas Emission Report Management issued by National Development and Reform Committee of P.R.China on 28/03/2021
20. Nameplate of the equipment

21. Meteorological data of Leiyang City:
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22. China's 14th Five-Year Plan and the 2035 Vision Outline
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23. Construction contract signed between the project owner and Leiyang Yongji construction engineering Co., Ltd on 28/08/2019
24. Equipment Purchase Agreement (solid-liquid separation system) signed between the project owner and Hunan Runzhaoda agricultural science and technology development Co., LTD on 10/09/2019
25. Equipment Purchase Agreement (Tilting and tossing machine) signed between the project owner and Zhengzhou Rui Yi constant heavy equipment Co., LTD
26. Public information about lifetime of organic fertilizer industry
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https://www.ndrc.gov.cn/fggz/gdzctz/tzfg/201907/t20190729_1197578.html
28. Business License of Leiyang Cailun Contemporary Agricultural Science and Technology Park Development Co., Ltd.
29. VCS monitoring manual
30. Local stakeholder consultation questionnaire
31. Regulations on Pollution Prevention Control of Livestock and Poultry Scale Farming in Hunan Province
http://www.hunan.gov.cn/hnszf/xgk/wjk/szfbgt/202209/t20220909_28733974.html
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33. Design Code for Wastewater Stabilization Ponds (GJJ/T54-93)
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34. Operation log covering this monitoring period
35. Announcement on deepening VAT reform related policies, issued by Ministry of Finance, General Administration of Taxation and Customs on 20/03/2019 (Announcement No.39 of 2019 by the General Administration of Customs of the Ministry of Finance)
<http://www.chinatax.gov.cn/chinatax/n810341/n810765/n4182981/201903/c4358850/content.html>
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https://www.ndrc.gov.cn/fggz/gdzctz/tzfg/201907/t20190729_1197578.html
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 39. Decision of The State Council on amending the Interim Provisions on levying Additional Education Fees
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 41. Income Tax Law of the People's Republic of China for Enterprises issued by the Chairman of the People's Republic of China
 42. Compilation method of municipal engineering investment estimation issued by Ministry of Construction of China
 43. Rules for the Provisional Regulations on Enterprise Income Tax of the People's Republic of China
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 44. 2021 Baseline Emission Factors for Regional Power Grids in China dated 17/11/2023 published by China DNA
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 45. Solid Liquid Separation Record covering this monitoring period
 46. Electricity Meter Reading Record covering this monitoring period
 47. Electricity Purchase Receipts covering this monitoring period
 48. Calibration Certificates of scale issued by Hengyang County Quality Inspection and Metrological Verification Center on 21/03/2022 valid till 20/03/2023, and issued on 21/03/2023 valid till 20/03/2024
 49. Calibration Accreditation of Hengyang County Quality Inspection and Metrological Verification Center, issued by Hunan Provincial Market Supervision Administration on 14/10/2019 valid till 13/10/2024
 50. Verification Regulation for Digital Indicating Weighing Instruments (JJG 539-2016)
<https://max.book118.com/html/2019/0615/7005032111002033.shtm>
 51. Settlement statement for completion of the first section issued by the project owner and construction party, i.e. Leiyang Yongji construction engineering Co., Ltd
 52. Settlement statement for completion of the second section issued by the project owner and construction party, i.e. Hunan Hengtai construction Engineering Co., Ltd

53. Settlement statement for completion of the third section issued by the project owner and construction party, i.e. Leiyang third construction Co., Ltd
54. Settlement statement for completion of the fourth section issued by the project owner and construction party, i.e. Leiyang fourth construction engineering Co., Ltd
55. Settlement statement for completion of the fifth section issued by the project owner and construction party, i.e. Hunan Runzhaoda agricultural science and technology development Co., Ltd
56. Calibration Certificates of analytical balance issued by Hengyang County Quality Inspection and Metrological Verification Center on 21/03/2022 valid till 20/03/2023, and issued on 21/03/2023 valid till 20/03/2024
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58. Published research article “Research progress on bio-fuel prepared from livestock manure by hydrothermal treatment”, finished by Zhang Shouyu Professor Research Group, School of Energy and Power Engineering, University of Shanghai for Science and Technology
59. Solid-liquid separation Records covering this monitoring period
60. Verification Regulation of Electronic Balance (JJG1036-2008)
https://wenku.baidu.com/view/bfb665fdd1d233d4b14e852458fb770bf78a3bc9.html?fr=sogou&_wkts_=1702971345736&needWelcomeRecommand=1
61. Organic Solid Fertilizers Sales Contract signed between Leiyang Chuangu food co., LTD and the project owner
62. Statement about O&M cost
63. Investment Decision Making Meeting Notice issued by the project owner on 15/04/2019
64. Announcement of the State Administration for Market Regulation on Issuing the Catalogue of Measuring Instruments for Compulsory Management” issued by the State Administration for Market Regulation on 23/10/2019
https://scj.sxxz.gov.cn/zljcssyzsfw/jlfw/202107/t20210726_3659463.html
65. Verification Regulation of Electrical Meters for Measuring Alternating-current Electrical Energy (JJG596-2012)
<https://www.doc88.com/p-8522868540952.html?r=1>
66. Employment Roster
67. List of livestock farms in Leiyang City where the swine manure was collected
68. 2019 Hunan Provincial Statistic Yearbook
<http://222.240.193.190/19tjni/indexch.htm>
69. Livestock Manure Treatment Agreement signed between the project owner and livestock farms in Leiyang City
70. Completion Acceptance Opinion issued by Hengyang Agriculture and Rural Bureau on

30/09/2022

71. Business Licences of Livestock Farms, from where swine were collected by the project activity
72. Organic Fertilizer Statistics
73. Grievance book
74. Statement about Raw Material of Organic Fertilizers issued by the project owner
75. Public information about manure accumulation time in lagoon before irrigation
https://www.nahs.org.cn/gk/tz/202211/t20221103_413061.htm
76. Historical photographs of baseline uncovered anaerobic lagoon
77. Statement about manure collected issued by the project owner
78. Equipment Maintenance Records since operation start date
79. Operation Records of Baseline Lagoons
80. Quality Control Plan made by PP in April 2022
81. Baseline manure transport records in 2020
82. Fertilizer Sales Record covering this monitoring period
83. Stock in and Out Transportation Records covering this monitoring period
84. Yearly Inventory Statistics covering this monitoring period
85. 2006 IPCC Report
<https://www.ipcc-nggip.iges.or.jp/public/2006gl/index.html>
86. Greenhouse Gas Emission Accounting Methods and Reporting Guidelines for Land Transportation Enterprises in China (Trial) issued by China NDRC on 06/07/2015
<https://www.ndrc.gov.cn/xxgk/zcfb/tz/201511/W020190905506438255108.pdf>
87. Statement about Swine Feed Purchase Statistics

APPENDIX 4: <CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)>

CL ID	01	Section no.	3.4.3	Date: 01/02/2024
Description of CL				
In section 3.3 of the joint PD& MR, all the emission sources of incremental CO ₂ emissions due to increased transportation project have been excluded for simplification since emission source is assumed to be very small. Please further explain why these emissions from these sources are assumed to be very small.				
Project proponent response				Date: 06/09/2024
In the project scenario, all separated solids undergo immediate on-site composting without storage, destruction or land application, prior to any secondary use. Therefore, the incremental transportation only incorporates itineraries for separated solids from the on-site solid separation system to the composting system. The transportation distance between these two phases is minimal, typically less than 0.05 km, and can be accomplished using wheelbarrows. Due to the negligible transportation distance, emissions from this source are assumed to be minimal and are consequently excluded for simplification.				
Documentation provided by project proponent				
updated joint PD&MR				
VVB assessment				Date: 10/09/2024
Via site inspection, it was verified the solids from the separation system immediately enter the composting system, and usually the separated solids are transported by using wheelbarrow, the itineraries where the transportation of separated solids occur was between separation system and composting system at the project site, which is less than 50m. Considering the short distance and the manual transport method, PP excluded emissions from transportation, which was accepted by the assessment team.				
The description has been supplemented in the updated joint PD&MR.				
Therefore, CL01 was closed.				

CAR ID	01	Section no.	3.3.3	Date: 01/02/2024
Description of CAR				
In section 2.3 of the joint PD&MR, please further supplement the measures taken to prevent				

environment impacts as per EIA Approval.	
Project proponent response	Date: 21/02/2024
Measures taken to prevent environment impacts in accordance with the EIA Approval have been illustrated in the joint PD&MR (version 1.1).	
Documentation provided by project proponent	
updated joint PD&MR	
VVB assessment	Date: 28/02/2024
The measures taken to prevent environment impacts have been supplemented in the updated joint PD&MR and verified to be consistent with that in the EIA Approval. Therefore, CAR01 was closed.	

CAR ID	02	Section no.	7.2	Date: 01/02/2024
Description of CAR				
In section 7.1 of the joint PD&MR, please supplement the data flow (including sampling), monitoring process and frequency of parameter $VS_{ss,y}$ and moisture content determine $M_{ss,y}$.				
Project proponent response				Date: 21/02/2024
Supplementary details regarding the data flow, monitoring processes, and frequency of parameter $VS_{ss,y}$ and moisture content (use for determining $M_{ss,y}$) have been provided in the joint PD&MR (version 1.1). In summary, Three samples are extracted daily from the separated solids for testing. The moisture content is initially determined using an oven, followed by the analysis of volatile matter using a muffle furnace. The final values for both parameters are obtained by averaging the test results from the three samples.				
Documentation provided by project proponent				
updated joint PD&MR				
VVB assessment				Date: 28/02/2024
The data flow (including sampling), monitoring process and frequency of parameter $VS_{ss,y}$ and moisture content determine $M_{ss,y}$ has been supplemented in the updated joint PD&MR and has been verified by the assessment team to be correct. Therefore, CAR02 was closed.				